

Eigenstate-Selective Entangled Two-Photon Absorption in Monolayer WSe₂

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We show that the Bell-state phase of a polarization-entangled photon pair controls the biexciton eigenstate distribution produced by entangled two-photon absorption (ETPA) in monolayer WSe₂. In a frequency-nondegenerate ladder scheme, two independent valley pathways (K and K') share no intermediate state, so the biphoton phase sets the relative amplitude between them. Within the valley-symmetric limit this phase factorizes from the material response, and the resulting selection rule partitions the excitation among biexciton eigenstates according to the Bell-state phase φ . The symmetric Bell state ($\varphi = 0$) selectively drives bright eigenstates, while the antisymmetric state ($\varphi = \pi$) drives the exchange-dark eigenstate. No classical polarization source reproduces this φ -dependent eigenstate distribution. Including valley dephasing and intervalley scattering at 4 K, the phase-scan visibility exceeds 0.97 for broadband SPDC ($T_e \sim 100$ fs) with high source purity.

I. INTRODUCTION

Entangled two-photon absorption (ETPA) refers to a two-photon absorption process driven by an entangled photon pair, typically generated by spontaneous parametric down-conversion (SPDC). Because the photons are entangled in time-energy, polarization, or both, ETPA is expected to differ qualitatively from classical two-photon absorption. The entangled signal scales linearly rather than quadratically with photon flux at low intensities, and the entanglement degrees of freedom can in principle be used to control the excitation pathway [1–4]. Experimental ETPA has been studied most extensively in molecular systems, where early fluorescence- and transmission-based measurements reported cross sections as large as $\sigma_e \sim 10^{-18}$ – 10^{-17} cm², suggesting a dramatic quantum advantage over classical excitation [5–7]. However, a series of recent experiments with carefully controlled systematics have placed upper bounds on σ_e that are orders of magnitude smaller, with many studies detecting no ETPA signal at all [8–10]. Several single-photon mechanisms that can mimic the linear flux dependence expected of ETPA have since been identified, including hot-band absorption from thermally populated vibrational levels [11], single-photon scattering [12], and linear optical losses in transmission geometries [13]. Because rate scaling alone cannot distinguish ETPA from these artifacts, the controversy motivates a search for qualitative entanglement signatures that have no classical counterpart. Such signatures have been explored theoretically, including time-energy entanglement that imposes pathway-selective interference in multilevel systems [14–17] and reveals classically dark bipolariton states in cavity-coupled molecular ensembles [4].

Monolayer transition metal dichalcogenides (TMDs) provide an attractive material platform for realizing such a qualitative signature, because their chiral optical selection rules directly couple photon polarization to the

valley degree of freedom of excited carriers. The valley structure originates from the inequivalent K and K' points of the hexagonal Brillouin zone, where circularly polarized light selectively couples to interband transitions in a single valley [18–21]. These materials host tightly bound excitons with binding energies of several hundred meV, arising from reduced dielectric screening and strong Coulomb interaction in two dimensions [22–24]. The chiral optical selection rule, combined with spin-valley locking due to the strong spin-orbit interaction, has enabled extensive studies of valley polarization, valley coherence, and valleytronic device concepts [25–27].

At the single-photon level, this polarization-to-valley coupling has been directly demonstrated. Tokman et al. [28] showed that absorbing a linearly polarized photon, a coherent superposition of σ^+ and σ^- in the circular basis, produces valley-entangled excitons in a TMD monolayer, with the entanglement verifiable through the polarization of the photoluminescence. This confirms that the polarization structure of quantum light can be coherently imprinted onto the valley degree of freedom.

Beyond single excitons, TMD monolayers support biexcitons (bound states of two excitons) with binding energies of order 20 meV [29–32]. The biexciton spectrum is significantly richer than that of the exciton, because electron–hole exchange interaction couples the six four-particle configurations that can be formed from excitons in the K and K' valleys into eigenstates with distinct symmetry and radiative properties [33]. Three eigenstates are optically bright and three are dark. Among the dark eigenstates, one (Φ_3) is especially relevant because it has nonzero overlap with the optically accessible cross-circular configurations, yet its cascade emission vanishes by destructive interference. This darkness originates not from a spin selection rule, as for the spin-forbidden dark exciton [34–36], but from the antisymmetric superposition structure of the eigenstate in configuration space. The fine structure was confirmed by ultrafast pump–probe spectroscopy [33]. Subsequent theoretical work has further explored the role of exchange in TMD exciton–exciton interactions [37, 38] and biexciton–polariton formation [39]. The question

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we address is whether the polarization-to-valley principle extends to two-photon absorption of an entangled photon pair, where the exchange-split biexciton eigenstate structure opens qualitatively new possibilities for state-selective control through the biphoton polarization phase. To our knowledge, no prior work has combined biphoton polarization entanglement with the exchange-split biexciton manifold of a TMD monolayer.

In this work, we propose such a symmetry-based ETPA observable, realized through the interplay between biphoton polarization entanglement and electron-hole exchange symmetry in the biexciton manifold of monolayer WSe₂. We consider a frequency-nondegenerate ladder scheme with the *A*-exciton as a real intermediate resonance, closely analogous to the entangled two-photon excitation demonstrated in atomic cesium [40] and rubidium [41], extended here to a solid-state setting. In Sec. II, we combine the valley optical selection rules with the exchange-split biexciton eigenstates of Steinhoff et al. [33] to derive eigenstate-resolved pumping rates and show that the Bell-state phase imposes selection rules that separately address the bright and dark eigenstate sectors. We then establish an upper bound on the dark-eigenstate pumping rate achievable by any separable source and show that no separable polarization state reproduces the entangled eigenstate distribution. Sec. III assesses the robustness of this selection rule against valley dephasing, intervalley scattering, and source-side imperfections at cryogenic temperatures.

II. THEORY: FROM ENTANGLED PHOTON PAIR TO EIGENSTATE POPULATION

A. Valley selection rules and two-photon amplitude

The total Hamiltonian for the coupled light-matter system is

$$H = H_{\text{mat}} + H_{\text{field}} + V, \quad (1)$$

where H_{mat} is the material Hamiltonian with eigenstates $\{|g\rangle, |X_K\rangle, |X_{K'}\rangle, |B_j\rangle\}$. Here $|g\rangle$ is the ground state, $|X_K\rangle$ and $|X_{K'}\rangle$ are the bright *A*-excitons in the *K* and *K'* valleys (created by σ^+ and σ^- photons, respectively), and $|B_j\rangle$ ($j = 1, \dots, 6$) are the six zero-momentum biexciton configurations classified by valley occupancy and spin pairing of the two constituent excitons [33] (Table I); H_{field} is the free electromagnetic field Hamiltonian; and V is the electric-dipole interaction in the rotating-wave approximation:

$$V = -\hat{\mathbf{d}} \cdot \hat{\mathbf{E}}^{(+)}(t) + \text{h.c.} \quad (2)$$

The positive-frequency electric field operator is decomposed into signal (*s*) and idler (*i*) modes with polarization σ [42]:

$$\hat{E}_\sigma^{(+)}(t) = \sum_{m \in \{s, i\}} \int d\omega \mathcal{E}_m(\omega) \hat{a}_{m, \sigma}(\omega) e^{-i\omega t}, \quad (3)$$

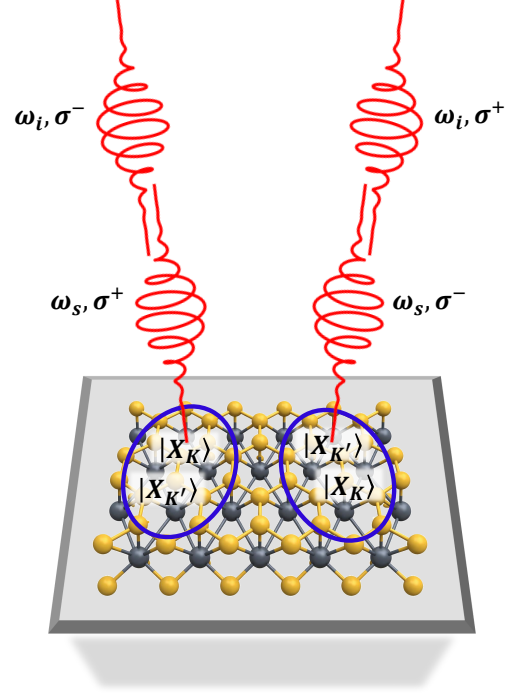


FIG. 1. Cross-circular two-photon excitation scheme in monolayer WSe₂. A polarization-entangled photon pair drives two valley pathways: (σ^+, σ^-) creates $|X_K\rangle$ then $|X_{K'}\rangle$ (left), while (σ^-, σ^+) creates $|X_{K'}\rangle$ then $|X_K\rangle$ (right).

where $\mathcal{E}_m(\omega)$ is the mode-envelope function and $\hat{a}_{m, \sigma}(\omega)$ annihilates a photon in mode *m* with polarization σ .

The biexciton is created by the absorption of two photons from the entangled pair. In the interaction picture with $H_0 = H_{\text{mat}} + H_{\text{field}}$, the time-evolution operator to second order in *V* is

$$U_I(t, -\infty) = \mathbb{I} + \left(-\frac{i}{\hbar}\right) \int_{-\infty}^t dt_1 V_I(t_1) + \left(-\frac{i}{\hbar}\right)^2 \int_{-\infty}^t dt_2 \int_{-\infty}^{t_2} dt_1 V_I(t_2) V_I(t_1) + \dots, \quad (4)$$

where $V_I(t) = e^{iH_0 t/\hbar} V e^{-iH_0 t/\hbar}$ and the time-ordering constraint $t_1 < t_2$ enforces causality. The second order is the leading contribution that connects the ground state to the biexciton manifold. Projecting the second-order term onto the final state $|B_j\rangle \otimes |0\rangle$ (biexciton with photon vacuum), the transition amplitude from the initial state $|g\rangle \otimes |\psi_{2\text{ph}}\rangle$ is

$$\mathcal{M}_j = \left(-\frac{i}{\hbar}\right)^2 \int_{-\infty}^t dt_2 \int_{-\infty}^{t_2} dt_1 \langle B_j | \langle 0 | V_I(t_2) V_I(t_1) | g \rangle |\psi_{2\text{ph}}\rangle, \quad (5)$$

where the two insertions of V_I account for the sequential absorption of two photons.

The input field state, $|\psi_{2\text{ph}}\rangle$, is a polarization-entangled photon pair generated by non-degenerate type-II spontaneous parametric down-conversion (SPDC) in a polarization-based Sagnac interferometer [43–45], shown schematically in Fig. 2, with a periodically poled KTP (ppKTP) crystal. The Sagnac output is $|\psi_{\text{lin}}\rangle = (|H_s V_i\rangle + e^{i\varphi} |V_s H_i\rangle)/\sqrt{2}$, where the Bell-state phase $\varphi = 4\theta_{\text{HWP1}}$ is tuned by a half-wave plate in the pump beam [45]. The non-degenerate signal and idler wavelengths are set by the phase-matching condition and tuned via the crystal temperature. The signal photon at frequency $\omega_s = E_X/\hbar$ ($\lambda_s \approx 713$ nm) is resonant with the A -exciton transition energy $E_X \approx 1.74$ eV [24]. The idler frequency is set by the target biexciton eigenstate. For the exchange-split fine structure of monolayer WSe_2 [33], the binding energies span from $E_B^{(\Phi_1)} \approx 19.8$ meV (singlet, brightest) through the triplet pair $E_B^{(\Phi_2)} \approx 12.3$ meV and $E_B^{(\Phi_3)} \approx 11.6$ meV, to $E_B^{(\Phi_6)} \approx 3.1$ meV (see Table I). Setting the fixed continuous-wave (CW) pump frequency to $\omega_p = \omega_s + \omega_i$ with $\omega_i = (E_X - E_B^{(\Phi_\alpha)})/\hbar$ places the sum frequency on resonance with eigenstate Φ_α . For example, targeting Φ_3 with $E_B \approx 11.6$ meV gives $\lambda_i \approx 718$ nm; targeting Φ_1 with $E_B \approx 19.8$ meV gives $\lambda_i \approx 721$ nm. A sufficiently narrow pump linewidth can therefore select a single eigenstate, whereas a broad linewidth may address multiple eigenstates simultaneously. A dichroic mirror and longpass filters remove the residual pump beam. Because a passive basis change from H - V to circular polarization would yield both cross- and co-circular components, waveplates in each arm (HWP3 + QWP5 in one arm and QWP4 in the other) implement the local maps $H \rightarrow \sigma^+$, $V \rightarrow \sigma^-$, with the HWP3 orientation selecting the cross-circular configuration [45]. This yields the cross-circular Bell state:

$$|\psi_{2\text{ph}}\rangle = \frac{1}{\sqrt{2}} \iint d\omega_s d\omega_i f(\omega_s, \omega_i) \times \left[\hat{a}_{s,\sigma^+}^\dagger(\omega_s) \hat{a}_{i,\sigma^-}^\dagger(\omega_i) + e^{i\varphi} \hat{a}_{s,\sigma^-}^\dagger(\omega_s) \hat{a}_{i,\sigma^+}^\dagger(\omega_i) \right] |0\rangle, \quad (6)$$

where $f(\omega_s, \omega_i)$ is the joint spectral amplitude (JSA) encoding the energy-time correlations of the SPDC output. Setting $\varphi = 0$ yields $|\psi^+\rangle$ and $\varphi = \pi$ yields $|\psi^-\rangle$.

The valley optical selection rules in monolayer WSe_2 dictate which intermediate exciton is created by each circular polarization [18–21]:

$$\sigma^+ \longrightarrow |X_K\rangle, \quad \sigma^- \longrightarrow |X_{K'}\rangle. \quad (7)$$

Explicitly, the nonvanishing dipole matrix elements are

$$\langle X_K | \hat{\mathbf{d}} \cdot \mathbf{e}_+ | g \rangle = d_0, \quad \langle X_{K'} | \hat{\mathbf{d}} \cdot \mathbf{e}_- | g \rangle = d_0, \quad (8)$$

with all cross-valley couplings vanishing ($\langle X_K | \hat{\mathbf{d}} \cdot \mathbf{e}_- | g \rangle = \langle X_{K'} | \hat{\mathbf{d}} \cdot \mathbf{e}_+ | g \rangle = 0$). Here $\mathbf{e}_\pm = \mp(\mathbf{e}_x \pm i\mathbf{e}_y)/\sqrt{2}$ are the circular polarization unit vectors and d_0 is the exciton oscillator strength.

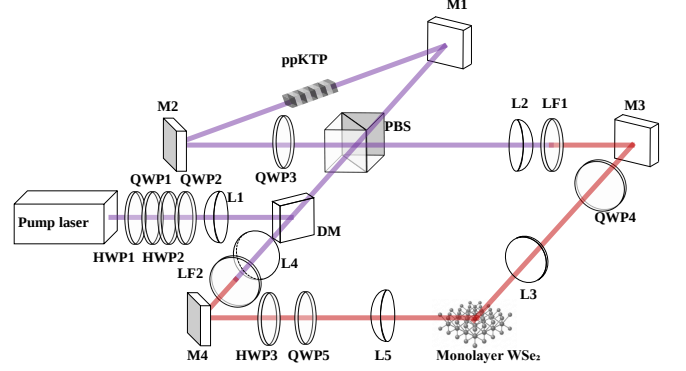


FIG. 2. Proposed experimental setup. A pump laser drives non-degenerate type-II SPDC in a Sagnac interferometer. HWP1 in the pump path sets the Bell-state phase φ . The DM and LFs remove the residual pump. Waveplates in each arm (QWP4 in one arm and HWP3 + QWP5 in the other) convert the H - V output to the cross-circular basis. PBS: polarizing beam splitter; DM: dichroic mirror; HWP1–HWP3: half-wave plates; QWP1–QWP5: quarter-wave plates; LF1, LF2: longpass filters; L1–L5: lenses; M1–M4: mirrors.

These selection rules, combined with the requirement that a biexciton consists of two excitons in opposite valleys with like spins, constrain the two-photon pathways to cross-circular polarization sequences [33]:

$$(\sigma^+, \sigma^-) : |g\rangle \xrightarrow{\omega_s, \sigma^+} |X_K\rangle \xrightarrow{\omega_i, \sigma^-} |B_2\rangle, \quad (9a)$$

$$(\sigma^-, \sigma^+) : |g\rangle \xrightarrow{\omega_s, \sigma^-} |X_{K'}\rangle \xrightarrow{\omega_i, \sigma^+} |B_3\rangle, \quad (9b)$$

where $|B_2\rangle$ and $|B_3\rangle$ are biexciton configurations composed of excitons in the (K, K') and (K', K) valley pairs, respectively. These are two of the six zero-momentum biexciton configurations classified by Steinhoff et al. [33]; the remaining four are inaccessible to cross-circular excitation (Table I lists all six configurations and their eigenstate decomposition). Each polarization sequence activates exactly one valley channel through exactly one intermediate state. This one-to-one mapping between polarization pair and biexciton configuration is the structural foundation of the eigenstate selectivity.

To evaluate the Dyson series amplitude Eq. (5), we insert a complete set of intermediate one-exciton states $\sum_n |n\rangle \langle n|$ between the two interactions. The matrix element becomes $\langle B_j | \hat{d}_{\sigma_2} | n \rangle \langle n | \hat{d}_{\sigma_1} | g \rangle$ (where subscripts 1 and 2 label the temporal vertex ordering, $t_1 < t_2$), which selects a specific valley intermediate $|n\rangle$ via the selection rules Eq. (7). The field part requires contracting the two annihilation operators $\hat{a}_{m_2, \sigma_2}(\omega_2) \hat{a}_{m_1, \sigma_1}(\omega_1)$ with the two creation operators $\hat{a}_{s, \mu}^\dagger(\omega_s) \hat{a}_{i, \nu}^\dagger(\omega_i)$ in $|\psi_{2\text{ph}}\rangle$ [Eq. (6)]. Since signal and idler occupy distinct modes with $[\hat{a}_{s, \mu}(\omega), \hat{a}_{i, \nu}^\dagger(\omega')] = 0$, there are exactly two ways to pair the annihilation operators with the creation operators.

tors.

The later interaction (t_2) annihilates the signal photon and the earlier interaction (t_1) annihilates the idler. The field matrix element evaluates to

$$\langle 0 | \hat{a}_{s,\sigma_2}(\omega_s) \hat{a}_{i,\sigma_1}(\omega_i) | \psi_{2\text{ph}} \rangle = \Psi(\omega_s, \sigma_2; \omega_i, \sigma_1). \quad (10)$$

which pairs the later vertex polarization σ_2 with the signal label and σ_1 with the idler label – Contraction A ($m_2 = s, m_1 = i$). The intermediate-state propagator is evaluated at the idler frequency ω_i .

The later interaction annihilates the idler photon and the earlier interaction annihilates the signal. The field matrix element evaluates to

$$\langle 0 | \hat{a}_{i,\sigma_2}(\omega_i) \hat{a}_{s,\sigma_1}(\omega_s) | \psi_{2\text{ph}} \rangle = \Psi(\omega_s, \sigma_1; \omega_i, \sigma_2). \quad (11)$$

with the polarization indices swapped relative to Contraction A – thus, Contraction B ($m_2 = i, m_1 = s$). The intermediate-state propagator is now evaluated at the signal frequency ω_s .

Evaluating the time-ordered integral in Eq. (5) (Appendix A), each contraction yields a retarded intermediate-state propagator and a common final-state Lorentzian $L(\omega_s + \omega_i) = [(\omega_{XX} - \omega_s - \omega_i) - i\gamma_B]^{-1}$, where $\omega_{XX} \equiv (E_{B_j} - E_g)/\hbar$ is the biexciton transition frequency from the ground state, enforcing two-photon energy conservation.

Defining the frequency-dependent two-photon matrix element

$$T_j(\omega) = \sum_n \frac{\langle B_j | \hat{d}_{\sigma_2} | n \rangle \langle n | \hat{d}_{\sigma_1} | g \rangle}{\hbar(\omega_{ng} - \omega) - i\hbar\gamma_n}, \quad (12)$$

the total amplitude for populating configuration $|B_j\rangle$ takes the form (see Appendix A)

$$\begin{aligned} \mathcal{M}_j = \iint d\omega_s d\omega_i & \left[\Psi(\omega_s, \sigma_2; \omega_i, \sigma_1) T_j(\omega_i) \right. \\ & \left. + \Psi(\omega_s, \sigma_1; \omega_i, \sigma_2) T_j(\omega_s) \right] L(\omega_s + \omega_i), \end{aligned} \quad (13)$$

where $\Psi(\omega_s, \sigma; \omega_i, \sigma')$ is the biphoton effective wavefunction for polarization pair (σ, σ') . Each term is weighted by a different joint spectral amplitude component, reflecting the polarization label of whichever photon is absorbed at each vertex.

Since only cross-circular polarization pairs contribute to the two-photon pathways [Eqs. (9a) and (9b)], the relevant components are those with $(\mu, \nu) \in \{(\sigma^+, \sigma^-), (\sigma^-, \sigma^+)\}$. Evaluating these from the explicit biphoton state [Eqs. (6), (10), and (11)]:

$$\Psi(\omega_s, \sigma^+; \omega_i, \sigma^-) = \frac{f(\omega_s, \omega_i)}{\sqrt{2}}, \quad (14a)$$

$$\Psi(\omega_s, \sigma^-; \omega_i, \sigma^+) = \frac{e^{i\varphi} f(\omega_s, \omega_i)}{\sqrt{2}}. \quad (14b)$$

The key sign relation follows immediately:

$$\Psi(\omega_s, \sigma^-; \omega_i, \sigma^+) = e^{i\varphi} \Psi(\omega_s, \sigma^+; \omega_i, \sigma^-). \quad (15)$$

For $\varphi = 0$ ($|\psi^+\rangle$), the two components are equal; for $\varphi = \pi$ ($|\psi^-\rangle$), they differ by a sign. The phase $e^{i\varphi}$ encodes the polarization correlation of the entangled pair, which is a distinct degree of freedom from the frequency correlations contained in f .

Substituting Eq. (14) into Eq. (13) for the two cross-circular configurations yields

$$\begin{aligned} \mathcal{M}_2 = \frac{1}{\sqrt{2}} \iint d\omega_s d\omega_i & f(\omega_s, \omega_i) \\ & \times \left[e^{i\varphi} T_2(\omega_i) + T_2(\omega_s) \right] L(\omega_s + \omega_i), \end{aligned} \quad (16a)$$

$$\begin{aligned} \mathcal{M}_3 = \frac{1}{\sqrt{2}} \iint d\omega_s d\omega_i & f(\omega_s, \omega_i) \\ & \times \left[T_3(\omega_i) + e^{i\varphi} T_3(\omega_s) \right] L(\omega_s + \omega_i). \end{aligned} \quad (16b)$$

The Bell-state phase $e^{i\varphi}$ appears on the $T_2(\omega_i)$ term in $\mathcal{M}_2$ but on the $T_3(\omega_s)$ term in $\mathcal{M}_3$, reflecting which polarization branch of the Bell state feeds each valley pathway.

B. Biexciton fine structure and eigenstate projection

The electron-hole exchange interaction mixes the six zero-momentum biexciton configurations $|B_1\rangle$ – $|B_6\rangle$ [33]. Diagonalizing the configuration-basis Hamiltonian – which includes nonlocal exchange (coupling constant U) and local-field corrections – yields six eigenstates whose complete decomposition in the configuration basis, energies, binding energies, and emission properties are collected in Table I.

Of the six eigenstates, four ($\Phi_1, \Phi_2, \Phi_3, \Phi_6$) have nonzero overlap with the optically accessible cross-circular configurations $|B_2\rangle$ and $|B_3\rangle$; the remaining two (Φ_4, Φ_5) are inaccessible in this sector. The symmetry column in Table I reveals the essential structural ingredient for the selection rule, namely that all three bright eigenstates have equal overlaps with $|B_2\rangle$ and $|B_3\rangle$ (symmetric under $B_2 \leftrightarrow B_3$), while Φ_3 is the unique antisymmetric eigenstate in the full six-dimensional Hilbert space. Its vanishing oscillator strength follows directly, $d_- \propto \langle \Phi_3 | B_2 \rangle + \langle \Phi_3 | B_3 \rangle = 0$.

1. Eigenstate decomposition of the cross-circular sector

The symmetric and antisymmetric superpositions of $|B_2\rangle$ and $|B_3\rangle$ have distinct eigenstate decompositions (Table I):

$$\frac{1}{\sqrt{2}}(|B_2\rangle + |B_3\rangle) = -\frac{1}{\sqrt{3}}|\Phi_1\rangle + \frac{1}{\sqrt{2}}|\Phi_2\rangle + \frac{1}{\sqrt{6}}|\Phi_6\rangle, \quad (17a)$$

$$\frac{1}{\sqrt{2}}(|B_2\rangle - |B_3\rangle) = |\Phi_3\rangle. \quad (17b)$$

TABLE I. Biexciton eigenstates from the exchange Hamiltonian [33]. Single ($\uparrow/\downarrow$) and double ($\uparrow\uparrow/\downarrow\downarrow$) arrows denote electron and hole spin projections, respectively, with $\uparrow/\uparrow$ residing at K and $\downarrow/\downarrow$ at K' ; each four-arrow symbol is ordered as $e_1h_1e_2h_2$ for the two electron-hole pairs composing the configuration. Columns list the expansion coefficients $\langle\Phi_\alpha|B_j\rangle$ in the configuration basis, the eigenenergy in the single-constant approximation (from Eq. (8) of Ref. 33), the binding energy E_B from the full microscopic calculation for WSe₂ on apphire [33], the $B_2 \leftrightarrow B_3$ symmetry class, the cross-circular oscillator strength $|d_\alpha|^2 \propto |\langle\Phi_\alpha|B_2\rangle + \langle\Phi_\alpha|B_3\rangle|^2$, and bright/dark character. The E_B values include local-field corrections.

Φ_α	B_1	B_2	B_3	B_4	B_5	B_6	Energy	E_B (meV)	$B_2 \leftrightarrow B_3$	$ d_\alpha ^2$	Character
	$\uparrow\uparrow\uparrow\uparrow$	$\uparrow\uparrow\downarrow\downarrow$	$\downarrow\downarrow\uparrow\uparrow$	$\uparrow\downarrow\downarrow\uparrow$	$\downarrow\uparrow\uparrow\downarrow$	$\downarrow\downarrow\downarrow\downarrow$					
Φ_1	$1/\sqrt{6}$	$-1/\sqrt{6}$	$-1/\sqrt{6}$	$-1/\sqrt{6}$	$-1/\sqrt{6}$	$1/\sqrt{6}$	E	19.8	Sym.	2/3	Bright
Φ_2	0	1/2	1/2	-1/2	-1/2	0	$E + 2U$	12.3	Sym.	1	Bright
Φ_3	0	$1/\sqrt{2}$	$-1/\sqrt{2}$	0	0	0	$E + 2U$	11.6	Anti.	0	Dark
Φ_4	0	0	0	$1/\sqrt{2}$	$-1/\sqrt{2}$	0	$E + 2U$	11.6	—	—	Dark
Φ_5	$1/\sqrt{2}$	0	0	0	0	$-1/\sqrt{2}$	$E + 4U$	3.4	—	—	Dark
Φ_6	$2/\sqrt{12}$	$1/\sqrt{12}$	$1/\sqrt{12}$	$1/\sqrt{12}$	$1/\sqrt{12}$	$2/\sqrt{12}$	$E + 6U$	3.1	Sym.	1/3	Bright

The symmetric superposition has nonzero overlap with all three bright eigenstates; the antisymmetric superposition is exactly the dark eigenstate Φ_3 . These overlaps determine which eigenstates are accessible for a given Bell phase and set the geometric coupling efficiency for each eigenstate, but the absolute excitation rate also depends on the spectral factor κ_α (Sec. II C).

2. Exact phase factorization

Time-reversal symmetry of the TMD band structure at zero magnetic field guarantees identical dipole matrix elements and dephasing rates in the two valleys ($d_0^{(K)} = d_0^{(K')}$, $\gamma_K = \gamma_{K'}$), so that the two-photon kernels satisfy $T_2(\omega) = T_3(\omega) \equiv T(\omega)$. Since the eigenstate decomposition [Eqs. (17a) and (17b)] shows that $\mathcal{M}_2 + \mathcal{M}_3$ controls the bright eigenstates (Φ_1, Φ_2, Φ_6) while $\mathcal{M}_2 - \mathcal{M}_3$ controls the dark eigenstate Φ_3 , we form these combinations from the cross-circular amplitudes [Eqs. (16a) and (16b)]:

$$\mathcal{M}_2 + \mathcal{M}_3 = \frac{1+e^{i\varphi}}{\sqrt{2}} \iint d\omega_s d\omega_i f [T(\omega_s) + T(\omega_i)] L, \quad (18a)$$

$$\mathcal{M}_2 - \mathcal{M}_3 = \frac{1-e^{i\varphi}}{\sqrt{2}} \iint d\omega_s d\omega_i f [T(\omega_s) - T(\omega_i)] L. \quad (18b)$$

where $f \equiv f(\omega_s, \omega_i)$ and $L \equiv L(\omega_s + \omega_i)$. The prefactors $(1 \pm e^{i\varphi})$ are independent of the spectral integrals; the Bell-state phase factorizes exactly from the material response.

For the symmetric Bell state $|\psi^+\rangle$ ($\varphi = 0$), the antisymmetric combination vanishes, $\mathcal{M}_2 - \mathcal{M}_3 = 0$, so the dark eigenstate Φ_3 , which requires precisely this combination Eq. (17b), receives zero population. The excitation is then confined to the bright sector.

For the antisymmetric Bell state $|\psi^-\rangle$ ($\varphi = \pi$), the symmetric combination vanishes, $\mathcal{M}_2 + \mathcal{M}_3 = 0$, so all

bright eigenstates, which require the symmetric combination Eq. (17a), receive zero population. The prepared state is then the antisymmetric eigenstate $|\Phi_3\rangle$.

These zeros rely on two independent ingredients, namely (i) valley symmetry, $T_2(\omega) = T_3(\omega)$, which makes the two cross-circular pathways share the same material response function; and (ii) the common-JSA assumption of Eq. (6), which ensures both Bell components carry the same spectral amplitude. Each ingredient can break independently. Corrections enter through (i) valley anisotropy (strain, magnetic field, or substrate-induced effects), which breaks the $T_2 = T_3$ identity, or (ii) polarization-dependent spectral distortion in the beam path (e.g., chromatic birefringence), which gives the two Bell components different JSAs.

3. Dominant-contraction approximation

A further simplification arises from the non-degenerate excitation scheme. Because the signal and idler central frequencies are separated by the binding energy $E_B^{(\Phi_\alpha)} \gg \hbar\gamma_X \approx 1.5$ meV [46, 47], only one time-ordering has a resonant intermediate state. The suppression ratio is

$$\frac{|T(\omega_i)|^2}{|T(\omega_s)|^2} = \frac{\hbar^2 \gamma_X^2}{(E_B^{(\Phi_\alpha)})^2 + \hbar^2 \gamma_X^2} \approx \left(\frac{\hbar\gamma_X}{E_B^{(\Phi_\alpha)}} \right)^2, \quad (19)$$

which is ≈ 0.016 for Φ_2/Φ_3 ($E_B \approx 12$ meV) and ≈ 0.006 for Φ_1 ($E_B \approx 19.8$ meV), so the off-resonant time-ordering is suppressed by roughly two orders of magnitude in rate. Dropping the off-resonant contraction ($T_2(\omega_i)$ in Eq. (16a) and $T_3(\omega_i)$ in Eq. (16b)) gives

$$\mathcal{M}_2 \approx \frac{1}{\sqrt{2}} \iint d\omega_s d\omega_i f T_2(\omega_s) L, \quad (20a)$$

$$\mathcal{M}_3 \approx \frac{e^{i\varphi}}{\sqrt{2}} \iint d\omega_s d\omega_i f T_3(\omega_s) L. \quad (20b)$$

The Bell-state phase $e^{i\varphi}$ survives only in $\mathcal{M}_3$, since in $\mathcal{M}_2$ it entered through the off-resonant contraction

$T_2(\omega_i)$, which has been suppressed. Valley symmetry ($T_2 = T_3$, Sec. II B 2) then yields the simple ratio

$$\mathcal{M}_3 \approx e^{i\varphi} \mathcal{M}_2, \quad (21)$$

valid to corrections of order $\gamma_X/E_B^{(\Phi_\alpha)}$ in amplitude.

Since cross-circular excitation can only populate $|B_2\rangle$ and $|B_3\rangle$ [Eqs. (9a) and (9b)], the second-order perturbation amplitude Eq. (5) produces a biexciton state proportional to the sum of configuration amplitudes:

$$|\psi_{\text{BX}}\rangle \propto \mathcal{M}_2 |B_2\rangle + \mathcal{M}_3 |B_3\rangle. \quad (22)$$

Substituting Eq. (21), the common scalar amplitude $\mathcal{M}_2$ factors out:

$$|\psi_{\text{BX}}\rangle \propto \mathcal{M}_2 (|B_2\rangle + e^{i\varphi} |B_3\rangle). \quad (23)$$

Because $\mathcal{M}_2$ is an overall complex number, it does not affect the quantum state and drops out upon normalization. The biexciton configurations are orthogonal, $\langle B_j | B_k \rangle = \delta_{jk}$, so $\| |B_2\rangle + e^{i\varphi} |B_3\rangle \|^2 = 1 + |e^{i\varphi}|^2 = 2$. The normalized prepared state is therefore

$$|\psi_{\text{prep}}\rangle = \frac{1}{\sqrt{2}} (|B_2\rangle + e^{i\varphi} |B_3\rangle), \quad (24)$$

which interpolates continuously between the symmetric ($\varphi = 0$) and antisymmetric ($\varphi = \pi$) limits of Eqs. (17a) and (17b). This state encodes the relative phase and amplitude between valley pathways as set by the Bell-state phase.

Within the dominant-contraction limit, valley symmetry guarantees that the K and K' pathways have identical material response ($T_2(\omega) = T_3(\omega)$). As a result, both configuration amplitudes share a common resonant two-photon amplitude $\mathcal{T}$ (see Appendix B):

$$\mathcal{M}_2 = \frac{2\pi \mathcal{N} L(\omega_p)}{\sqrt{2} T_e} \mathcal{T}, \quad \mathcal{M}_3 = \frac{2\pi e^{i\varphi} \mathcal{N} L(\omega_p)}{\sqrt{2} T_e} \mathcal{T}, \quad (25)$$

where $\mathcal{N}$ is the JSA normalization constant (Appendix B, Eq. (B1)), $L(\omega_p)$ is the final-state Lorentzian, and

$$\mathcal{T} = \frac{d_0 \cdot d_{XX,X}}{-i\hbar\gamma_X} \left(1 - e^{-\gamma_X T_e/2} \right). \quad (26)$$

Here $d_{XX,X}$ is the exciton-to-biexciton transition dipole, T_e is the entanglement time of the biphoton wavepacket, and γ_X is the exciton dephasing rate. Eq. (26) is the result of evaluating the spectral integral in the dominant-contraction amplitude for the physical sinc-shaped JSA of type-II SPDC.

The complete T_e dependence of the transition amplitude Eq. (25) resides in the factor $(1 - e^{-\gamma_X T_e/2})/T_e$, which encodes the spectral overlap between the biphoton bandwidth and the exciton resonance. In the broad-bandwidth limit ($\gamma_X T_e \ll 1$), only a small spectral fraction overlaps with the resonance; the overlap factor reduces to $\gamma_X T_e/2$, canceling the $1/(\gamma_X T_e)$ dependence

so that the amplitude and pumping rate saturate to constants independent of T_e and γ_X . In the narrow-bandwidth limit ($\gamma_X T_e \gg 1$), the overlap saturates to unity but the amplitude still decreases as $1/T_e$, so further narrowing only reduces the pumping rate without improving resonance coverage.

C. Pumping rates

The transition probability into eigenstate $|\Phi_\alpha\rangle$ is obtained by projecting the prepared biexciton state onto $|\Phi_\alpha\rangle$. For a general input light state, the pumping rate is (see Appendix C for derivation)

$$W_\alpha = \kappa_\alpha \sum_{\mu\nu, \mu'\nu'} \langle \Phi_\alpha | B_{\mu\nu} \rangle \langle B_{\mu'\nu'} | \Phi_\alpha \rangle \rho_{\mu\nu, \mu'\nu'}, \quad (27)$$

where $B_{\mu\nu}$ denotes the configuration activated by polarization pair (μ, ν) , with $B_{\sigma^+\sigma^-} = B_2$, $B_{\sigma^-\sigma^+} = B_3$ [Eqs. (9a) and (9b)], and analogously for the co-circular pairs. The matrix $\rho_{\mu\nu, \mu'\nu'}$ is the polarization-sector density matrix of the input biphoton state. The eigenstate-dependent prefactor is

$$\kappa_\alpha \equiv \kappa_0 \frac{\gamma_B^2}{(\omega_{\Phi_{\alpha g}} - \omega_p)^2 + \gamma_B^2}, \quad (28)$$

where

$$\kappa_0 \equiv R_{\text{pairs}} \frac{2\pi^2 |\mathcal{N}|^2 |\mathcal{T}|^2}{T_e^2 \gamma_B^2} \quad (29)$$

is the on-resonance pumping-rate prefactor (i.e., the value of κ_α when $\omega_p = \omega_{\Phi_{\alpha g}}$), R_{pairs} is the photon pair flux, $\mathcal{N}$ is the joint-spectral-amplitude normalization constant [Eq. (B1)], and γ_B is the biexciton linewidth (half-width at half-maximum). In the equal-resonance limit where all eigenstates share the same Lorentzian weight, $\kappa_\alpha \rightarrow \kappa_0$. The linear scaling $W_\alpha \propto R_{\text{pairs}}$ is a signature of ETPA [3, 16, 40].

We now evaluate Eq. (27) for the continuously tunable Bell state Eq. (6). Because the spectral degrees of freedom have already been absorbed into the state-dependent prefactors κ_α , the remaining light-state dependence enters only through the polarization-sector density matrix $\rho_{\mu\nu, \mu'\nu'}$. For a general biphoton polarization state $|\psi\rangle_{\text{Bell}} = \sum_{\mu\nu} c_{\mu\nu} |\mu\rangle_s |\nu\rangle_i$, this matrix has diagonal elements (populations) $p_{\mu\nu} \equiv \langle \psi | \hat{a}_{s,\mu}^\dagger \hat{a}_{i,\nu}^\dagger \hat{a}_{i,\nu} \hat{a}_{s,\mu} | \psi \rangle = |c_{\mu\nu}|^2$ and off-diagonal elements (coherences) $\rho_{\mu\nu, \mu'\nu'} \equiv \langle \psi | \hat{a}_{s,\mu}^\dagger \hat{a}_{i,\nu}^\dagger \hat{a}_{i,\nu'} \hat{a}_{s,\mu'} | \psi \rangle = c_{\mu\nu} c_{\mu'\nu'}^*$. For the biphoton state $|\psi_{2\text{ph}}\rangle$ of Eq. (6), the polarization state is the cross-circular Bell state

$$|\psi(\varphi)\rangle_{\text{Bell}} = \frac{1}{\sqrt{2}} (|\sigma^+\rangle_s |\sigma^-\rangle_i + e^{i\varphi} |\sigma^-\rangle_s |\sigma^+\rangle_i), \quad (30)$$

with polarization density matrix

$$\rho_{\text{Bell}}(\varphi) = |\psi(\varphi)\rangle_{\text{Bell}} \langle \psi(\varphi)|_{\text{Bell}} \quad (31)$$

and nonzero matrix elements

$$p_{\sigma^+\sigma^-} = p_{\sigma^-\sigma^+} = \frac{1}{2}, \quad \rho_{\sigma^+\sigma^-, \sigma^-\sigma^+} = \frac{e^{-i\varphi}}{2}.$$

In the cross-circular sector, only the $(\mu, \nu) \in \{(\sigma^+, \sigma^-), (\sigma^-, \sigma^+)\}$ pairs contribute. For each eigenstate, the sum in Eq. (27) reduces to four terms:

$$\begin{aligned} W_\alpha = \kappa_\alpha & \left[|\langle \Phi_\alpha | B_2 \rangle|^2 p_{\sigma^+\sigma^-} + |\langle \Phi_\alpha | B_3 \rangle|^2 p_{\sigma^-\sigma^+} \right. \\ & + \langle \Phi_\alpha | B_2 \rangle \langle B_3 | \Phi_\alpha \rangle \rho_{\sigma^+\sigma^-, \sigma^-\sigma^+} \\ & \left. + \langle \Phi_\alpha | B_3 \rangle \langle B_2 | \Phi_\alpha \rangle \rho_{\sigma^-\sigma^+, \sigma^+\sigma^-} \right]. \end{aligned} \quad (32)$$

For the pure Bell state, this four-term sum simplifies to a compact form. Using $p_{\sigma^+\sigma^-} = p_{\sigma^-\sigma^+} = 1/2$ and $\rho_{\sigma^+\sigma^-, \sigma^-\sigma^+} = e^{-i\varphi}/2$, the four terms in Eq. (32) combine into

$$W_\alpha(\varphi) = \kappa_\alpha |\langle \Phi_\alpha | \psi_{\text{prep}} \rangle|^2, \quad (33)$$

where $|\psi_{\text{prep}}\rangle = (|B_2\rangle + e^{i\varphi}|B_3\rangle)/\sqrt{2}$ [Eq. (24)]. This step uses the fact that the overlaps $\langle \Phi_\alpha | B_2 \rangle$ and $\langle \Phi_\alpha | B_3 \rangle$ in Table I are all real (the exchange Hamiltonian in the configuration basis is a real symmetric matrix), so the cross terms in Eq. (32) combine straightforwardly with the diagonal terms into the squared modulus. The separation of W_α into a spectral prefactor κ_α and a geometric overlap makes the symmetry-based selection rule transparent. The φ dependence resides entirely in the overlap, while the spectral factor κ_α is phase-independent.

We now evaluate the overlaps using the eigenvectors classified in Table I. For each bright eigenstate $\alpha \in \{1, 2, 6\}$, the symmetric overlaps $\langle \Phi_\alpha | B_2 \rangle = \langle \Phi_\alpha | B_3 \rangle \equiv g_\alpha$ (with $g_1 = -1/\sqrt{6}$, $g_2 = 1/2$, $g_6 = 1/\sqrt{12}$) give

$$\begin{aligned} W_\alpha(\varphi) &= \kappa_\alpha |g_\alpha|^2 [p_{\sigma^+\sigma^-} + p_{\sigma^-\sigma^+} + 2\text{Re}(\rho_{\sigma^+\sigma^-, \sigma^-\sigma^+})] \\ &= \kappa_\alpha |g_\alpha|^2 (1 + \cos \varphi), \quad \alpha \in \{1, 2, 6\}. \end{aligned} \quad (34)$$

Every bright eigenstate individually carries the same $(1 + \cos \varphi)$ modulation; only the overall scale $\kappa_\alpha |g_\alpha|^2$ differs. For the dark eigenstate (the antisymmetric superposition $|B_2\rangle - |B_3\rangle$), the cross term $\langle \Phi_3 | B_2 \rangle \langle B_3 | \Phi_3 \rangle = -1/2$ reverses the coherence contribution, replacing $(1 + \cos \varphi)$ with $(1 - \cos \varphi)$:

$$\begin{aligned} W_{\Phi_3}(\varphi) &= \frac{\kappa_3}{2} [p_{\sigma^+\sigma^-} + p_{\sigma^-\sigma^+} - 2\text{Re}(\rho_{\sigma^+\sigma^-, \sigma^-\sigma^+})] \\ &= \frac{\kappa_3}{2} (1 - \cos \varphi). \end{aligned} \quad (35)$$

Figure 3(a) shows how the Bell-state phase redistributes excitation between the bright and dark symmetry sectors. At $\varphi = 0$ ($|\psi^+\rangle$), all excitation goes to the bright sector, with the eigenstate-resolved weights

$$W_1(0) = \frac{1}{3}\kappa_1, \quad W_2(0) = \frac{1}{2}\kappa_2, \quad W_6(0) = \frac{1}{6}\kappa_6. \quad (36)$$

The prefactors $1/3$, $1/2$, and $1/6$ are fixed geometric coupling efficiencies (the squared overlaps $|g_\alpha|^2 = |\langle \Phi_\alpha | B_2 \rangle|^2$) that quantify how strongly each eigenstate couples to the cross-circular sector. When the pump targets a specific eigenstate Φ_α , the spectral factor κ_α [Eq. (33)] is strongly peaked for that eigenstate, and the observed rate is $W_\alpha(0) = |g_\alpha|^2 \kappa_\alpha$. For example, targeting Φ_1 with $\omega_p \approx \omega_{\Phi_1, g}$ gives a rate $W_1(0) = \frac{1}{3}\kappa_1$, where the factor $1/3$ is the intrinsic geometric penalty for coupling to Φ_1 .

At $\varphi = \pi$ ($|\psi^-\rangle$), by contrast, every bright eigenstate vanishes individually. For every $\alpha \in \{1, 2, 6\}$:

$$W_\alpha(\pi) = \kappa_\alpha \times |\langle \Phi_\alpha | \psi_{\text{prep}}(\pi) \rangle|^2 = 0, \quad (37)$$

so the cascade emission from the targeted bright eigenstate vanishes at $\varphi = \pi$ within the model. Because the zero originates from the overlap $|\langle \Phi_\alpha | \psi_{\text{prep}}(\pi) \rangle|^2 = 0$, it holds for every bright eigenstate independently of κ_α . This zero holds beyond the dominant-contraction limit, as established by the exact factorization of Sec. II B 2. This symmetry-based extinction differs in mechanism from spectral selectivity. Detuning the pump frequency away from a bright eigenstate suppresses its rate through the Lorentzian tail of κ_α , and can achieve excellent contrast when eigenstates are spectrally well resolved relative to their linewidths. The Bell-phase mechanism, by contrast, produces a zero that is enforced by the exchange symmetry within the model and does not rely on the spectral separation of eigenstates.

For the dark eigenstate at $\varphi = \pi$, $W_3(\pi) = \kappa_3 \neq 0$ (as $|\langle \Phi_3 | \psi_{\text{prep}}(\pi) \rangle|^2 = 1$), but since Φ_3 is dark it produces no cascade emission. When the pump is tuned to Φ_3 ($E_B \approx 11.6$ meV), the Bell state at $\varphi = \pi$ channels all excitation into Φ_3 at its maximum rate, with bright-state contributions vanishing by the symmetry-based extinction alone, as shown in Fig. 3(b). Conversely, at $\varphi = 0$ under the same pump tuning, Φ_3 is suppressed by the symmetry-based extinction while Φ_1 and Φ_6 are suppressed by spectral detuning (~ 8 meV from the pump); only Φ_2 retains an appreciable rate because its 0.7-meV splitting from Φ_3 is comparable to the exciton linewidth $\hbar\gamma_X \approx 1.5$ meV, as seen in Fig. 3(b).

D. Comparison with classical polarization states

The pumping-rate modulation $W_\alpha(\varphi) \propto 1 \pm \cos \varphi$ for bright and dark cross-circular eigenstates [Eqs. (34)–(35)] originates from the biphoton cross-circular coherence $\rho_{\sigma^+\sigma^-, \sigma^-\sigma^+}$. A separable source with nonzero $\rho_{\sigma^+\sigma^-, \sigma^-\sigma^+}$ can also drive a phase-dependent eigenstate distribution, but we show below that the Peres–Horodecki separability criterion forces any such classical source to carry a compensating co-circular flux, which in turn modifies the eigenstate distribution at the Φ_2 , Φ_3 , Φ_5 , and Φ_6 sectors in ways that the Bell state avoids.

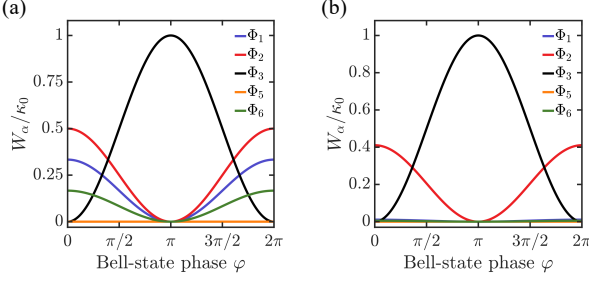


FIG. 3. Bell-state eigenstate-resolved pumping rates W_α/κ_0 . (a) Equal-resonance limit ($\kappa_\alpha = \kappa_0$ for all α): $\varphi = \pi$ extinguishes all bright eigenstates and channels the full excitation into the exchange-dark eigenstate Φ_3 ; $\varphi = 0$ suppresses Φ_3 completely. (b) With pump tuned to Φ_3 ($E_B \approx 11.6$ meV): Φ_3 dominates at $\varphi = \pi$, while residual bright-state rates are suppressed by both the symmetry-based extinction and spectral detuning.

1. General classical bound

A separable (classical) polarization state can be decomposed as an incoherent mixture of product states, $\rho_{\text{sep}} = \sum_k q_k \rho_s^{(k)} \otimes \rho_i^{(k)}$ ($q_k \geq 0$, $\sum_k q_k = 1$). Any such state has positive partial transpose (PPT) [48, 49]. Applied to the 2×2 principal submatrix of ρ^{T_i} indexed by $\{|\sigma^+\sigma^+\rangle, |\sigma^-\sigma^-\rangle\}$, PPT requires that its determinant be non-negative, giving

$$|\rho_{\sigma^+\sigma^-, \sigma^-\sigma^+}|^2 \leq p_{\sigma^+\sigma^+} p_{\sigma^-\sigma^-}, \quad (38)$$

with equality only for pure product states. For two qubits PPT is both necessary and sufficient for separability [49], so Eq. (38) holds for every classical source and any violation certifies polarization entanglement. Taking the square root of Eq. (38) and applying the inequality $\sqrt{p_{\sigma^+\sigma^+} p_{\sigma^-\sigma^-}} \leq (p_{\sigma^+\sigma^+} + p_{\sigma^-\sigma^-})/2$ yields

$$2|\rho_{\sigma^+\sigma^-, \sigma^-\sigma^+}| \leq p_{\sigma^+\sigma^+} + p_{\sigma^-\sigma^-} \equiv p_{\text{co}}, \quad (39)$$

so any phase-tunable classical source carries co-circular flux of at least twice its cross-circular coherence. The cross-circular Bell state Eq. (30) has $|\rho_{\sigma^+\sigma^-, \sigma^-\sigma^+}| = 1/2$ and $p_{\text{co}} = 0$, violating Eq. (39) by the largest possible margin.

2. Eigenstate-distribution consequences

The co-circular analog of Eq. (27) follows from restricting the polarization sum to the same-helicity pairs $(\sigma^+, \sigma^+) \rightarrow |B_1\rangle$ and $(\sigma^-, \sigma^-) \rightarrow |B_6\rangle$:

$$W_\alpha^{(\text{co})} = \kappa_\alpha^{(\text{co})} \sum_{(\mu\nu), (\mu'\nu')} \langle \Phi_\alpha | B_{\mu\nu} \rangle \langle B_{\mu'\nu'} | \Phi_\alpha \rangle \rho_{\mu\nu, \mu'\nu'}, \quad (40)$$

where $(\mu\nu), (\mu'\nu') \in \{(\sigma^+\sigma^+), (\sigma^-\sigma^-)\}$ with co-circular populations $p_{\sigma^+\sigma^+}$, $p_{\sigma^-\sigma^-}$ and coherence $\rho_{\sigma^+\sigma^+, \sigma^-\sigma^-}$.

We define the co-circular coupling efficiency $h_\alpha \equiv \langle \Phi_\alpha | B_1 \rangle = \langle \Phi_\alpha | B_6 \rangle$ for eigenstates symmetric under $B_1 \leftrightarrow B_6$, analogous to g_α in the cross-circular sector.

The Bell state concentrates all excitation on Φ_3 at $\varphi = \pi$, giving the dark-state rate $W_3^{(\text{Bell})}(\varphi = \pi) = \kappa_3^{(\text{cross})}$ (Sec. II B 2). We now ask how closely a classical source can match this rate. For any classical source, Eq. (35) generalizes to $W_3^{(\text{sep})} = (\kappa_3^{(\text{cross})}/2)[p_{\sigma^+\sigma^-} + p_{\sigma^-\sigma^+} - 2\text{Re}(\rho_{\sigma^+\sigma^-, \sigma^-\sigma^+})]$. Since $-\text{Re}(z) \leq |z|$ for any complex z , the bracketed quantity satisfies $[p_{\sigma^+\sigma^-} + p_{\sigma^-\sigma^+} - 2\text{Re}(\rho_{\sigma^+\sigma^-, \sigma^-\sigma^+})] \leq p_{\sigma^+\sigma^-} + p_{\sigma^-\sigma^+} + 2|\rho_{\sigma^+\sigma^-, \sigma^-\sigma^+}|$, and Eq. (39) together with the density-matrix normalization gives $p_{\sigma^+\sigma^-} + p_{\sigma^-\sigma^+} + 2|\rho_{\sigma^+\sigma^-, \sigma^-\sigma^+}| \leq p_{\sigma^+\sigma^+} + p_{\sigma^-\sigma^-} + p_{\sigma^+\sigma^+} + p_{\sigma^-\sigma^-} = 1$. The classical Φ_3 pumping rate is therefore bounded by

$$W_3^{(\text{sep})} \leq \frac{\kappa_3^{(\text{cross})}}{2}, \quad W_3^{(\text{Bell})}(\varphi = \pi) = \kappa_3^{(\text{cross})}, \quad (41)$$

so the Bell state achieves twice the maximum Φ_3 pumping rate of any classical source. The classical bound is saturated by a classical source for which $\text{Re}(\rho_{\sigma^+\sigma^-, \sigma^-\sigma^+}) = -|\rho_{\sigma^+\sigma^-, \sigma^-\sigma^+}| = -1/4$ and populations $p_{\sigma^+\sigma^-} = p_{\sigma^-\sigma^+} = p_{\sigma^+\sigma^+} = p_{\sigma^-\sigma^-} = 1/4$; at this saturating point, the bright cross-circular sector is simultaneously extinguished. At the saturating point, the co-circular antisymmetric eigenstate $\Phi_5 = (|B_1\rangle - |B_6\rangle)/\sqrt{2}$ (with $|h_5|^2 = 1/2$ and zero cross-circular overlap) is populated because $p_{\sigma^+\sigma^+} + p_{\sigma^-\sigma^-} = 1/2 > 0$, whereas the Bell state gives $W_5^{(\text{Bell})} = 0$ for every φ . A nonzero W_5 concurrent with a high Φ_3 rate therefore distinguishes classical from entangled sources.

To visualize these bounds concretely we take the balanced product state

$$|\psi(\theta)\rangle_{\text{sep}} = \left(\frac{|\sigma^+\rangle + |\sigma^-\rangle}{\sqrt{2}} \right)_s \otimes \left(\frac{|\sigma^+\rangle + e^{i\theta} |\sigma^-\rangle}{\sqrt{2}} \right)_i \quad (42)$$

as the classical reference, with populations $p_{\sigma^+\sigma^+} = p_{\sigma^+\sigma^-} = p_{\sigma^-\sigma^+} = p_{\sigma^-\sigma^-} = 1/4$ and coherences $\rho_{\sigma^+\sigma^-, \sigma^-\sigma^+} = e^{i\theta}/4$, $\rho_{\sigma^+\sigma^+, \sigma^-\sigma^-} = e^{-i\theta}/4$. Eq. (42) saturates both Eq. (38) [$|\rho_{\sigma^+\sigma^-, \sigma^-\sigma^+}|^2 = 1/16 = p_{\sigma^+\sigma^+} p_{\sigma^-\sigma^-}$] and, at $\theta = \pi$, the classical bound Eq. (41).

Because the Bell state drives each eigenstate only through the cross-circular overlap $|g_\alpha|^2$, eigenstates with dominant co-circular coupling are suppressed relative to the classical reference. For Φ_6 , with $|h_6|^2 = 1/3$ and $|g_6|^2 = 1/12$, the Bell rate $W_6^{(\text{Bell})} = \kappa_6^{(\text{cross})}(1 + \cos \varphi)/12$ carries no co-circular contribution, whereas the co-circular channel is the dominant contribution to the classical rate $W_6^{(\text{sep})} = (1 + \cos \theta)[\kappa_6^{(\text{cross})}/24 + \kappa_6^{(\text{co})}/6]$. For Φ_1 ($|g_1|^2 = |h_1|^2 = 1/6$), the loss of the co-circular pathways is exactly compensated by the doubled cross-circular flux of the Bell state, so $W_1^{(\text{Bell})}$ and $W_1^{(\text{sep})}$ coincide at matched phase. The eigenstate-resolved rates for both sources are collected in Table II. In the equal-resonance limit, Fig. 4(a) shows that $\theta = \pi$ channels excitation into Φ_3 at half the Bell rate of Fig. 3(a) and

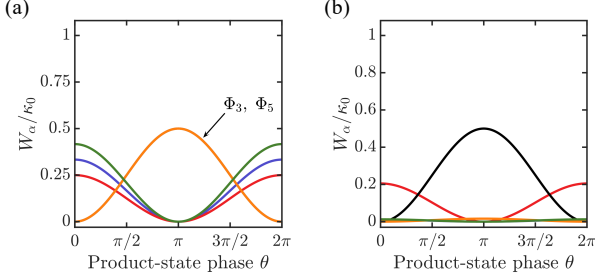


FIG. 4. Eigenstate-resolved pumping rates W_α/κ_0 for the classical reference of Eq. (42). (a) Equal-resonance limit: the Φ_3 peak is halved relative to the Bell state [Fig. 3(a)], and Φ_5 acquires nonzero population from the co-circular flux required by Eq. (39). (b) With pump tuned to Φ_3 ($E_B \approx 11.6$ meV): the peak Φ_3 rate is half that of Fig. 3(b) [Eq. (41)], and a residual Φ_5 background is visible.

into Φ_5 with equal weight, while Φ_6 receives a co-circular enhancement in the classical case that the Bell state does not access. With the pump tuned to Φ_3 , as illustrated in Fig. 4(b), the spectral factor $\kappa_3^{(\text{cross})}$ peaks while the other eigenstates are suppressed by spectral detuning, reproducing the dark-state selectivity of Fig. 3(b) but with the Φ_3 peak capped at half the Bell value [Eq. (41)] and a residual Φ_5 background characteristic of classical sources.

TABLE II. Eigenstate-resolved pumping rates for the cross-circular Bell state $|\psi(\varphi)\rangle_{\text{Bell}}$ [Eq. (30)] and the classical reference state $|\psi(\theta)\rangle_{\text{sep}}$ [Eq. (42)]. Coupling efficiencies $|g_\alpha|^2$ and $|h_\alpha|^2$ follow from the eigenvectors in Table I.

Φ_α	Bell: $W_\alpha^{(\text{Bell})}(\varphi)$	Classical: $W_\alpha^{(\text{sep})}(\theta)$
Φ_1	$\frac{\kappa_1^{(\text{cross})}}{6}(1 + \cos \varphi)$	$(1 + \cos \theta) \left[\frac{\kappa_1^{(\text{cross})}}{12} + \frac{\kappa_1^{(\text{co})}}{12} \right]$
Φ_2	$\frac{\kappa_2^{(\text{cross})}}{4}(1 + \cos \varphi)$	$\frac{\kappa_2^{(\text{cross})}}{8}(1 + \cos \theta)$
Φ_3	$\frac{\kappa_3^{(\text{cross})}}{2}(1 - \cos \varphi)$	$\frac{\kappa_3^{(\text{cross})}}{4}(1 - \cos \theta)$
Φ_4	0	0
Φ_5	0	$\frac{\kappa_5^{(\text{co})}}{4}(1 - \cos \theta)$
Φ_6	$\frac{\kappa_6^{(\text{cross})}}{12}(1 + \cos \varphi)$	$(1 + \cos \theta) \left[\frac{\kappa_6^{(\text{cross})}}{24} + \frac{\kappa_6^{(\text{co})}}{6} \right]$

III. DECOHERENCE AND VISIBILITY

In a real TMD monolayer, the intermediate exciton interacts with its environment before the second photon completes the biexciton transition. In the unsaturated regime ($\langle n \rangle \sim 10^{-4}$ – 10^{-7} per mode) [16, 50–52], the observable cascade emission intensity is proportional to the pumping rate, $I(\varphi) \propto W_\alpha(\varphi)$. We quantify the robust-

ness of the selection rule through the phase-scan visibility

$$V_{\text{total}} \equiv \frac{I(0) - I(\pi)}{I(0) + I(\pi)}, \quad (43)$$

which measures the depth of the selection-rule extinction at $\varphi = \pi$. A near-unit value confirms that the exchange-symmetry-based extinction survives phonon decoherence and is experimentally resolvable. We first show that the Bell-state phase is protected by material-idler entanglement and identify which-valley leakage as the sole visibility-degrading material-side mechanism (Sec. III A). We then quantify the resulting visibility loss from valley dephasing and intervalley scattering (Sec. III B), dark-to-bright relaxation within the triplet pair (Sec. III C), and polarization-dependent loss and partial entanglement on the source side (Sec. III D).

A. Entanglement-protected phase transfer

After the first photon is absorbed from the Bell state, the joint state of (material exciton + remaining idler photon) is

$$|\psi_{X,i}\rangle = \frac{1}{\sqrt{2}}(|X_K\rangle \otimes |\sigma^-\rangle_i + e^{i\varphi} |X_{K'}\rangle \otimes |\sigma^+\rangle_i). \quad (44)$$

Because the two terms are correlated with orthogonal idler polarizations, the reduced material state $\rho_{\text{mat}} = \text{Tr}_i |\psi_{X,i}\rangle \langle \psi_{X,i}| = (|X_K\rangle \langle X_K| + |X_{K'}\rangle \langle X_{K'}|)/2$ carries no inter-valley coherence; the phase φ resides entirely in the material-idler entanglement. When the idler is subsequently absorbed, the valley selection rules [Eq. (7)] map $|X_K\rangle + \sigma^- \rightarrow |B_2\rangle$ and $|X_{K'}\rangle + \sigma^+ \rightarrow |B_3\rangle$, transferring φ into the off-diagonal coherence between the two biexciton configurations, $\langle B_2 | \rho | B_3 \rangle \propto e^{-i\varphi}$, that determines eigenstate selectivity. By contrast, single-photon valley excitation [28] with a linearly polarized photon $(|\sigma^+\rangle + |\sigma^-\rangle)/\sqrt{2}$ creates the valley-coherent superposition $(|X_K\rangle + |X_{K'}\rangle)/\sqrt{2}$, placing the coherence directly in ρ_{mat} where it is exposed to phonon dephasing on the timescale $\tau_{\text{valley}} \sim 1$ –3 ps.

Symmetric exciton damping, the homogeneous dephasing γ_X that enters the K - and K' -valley amplitudes $\mathcal{M}_2$ and $\mathcal{M}_3$ identically through Eq. (26), reduces the overall excitation rate but preserves the ratio $\mathcal{M}_3/\mathcal{M}_2 = e^{i\varphi}$ by valley symmetry and therefore does not degrade the eigenstate selectivity.

Within the model considered here, the primary mechanism that degrades the visibility is which-valley information leakage. If the phonon environment, initially in a common state $|\mathcal{E}_0\rangle$, evolves into valley-dependent states $|\mathcal{E}_K\rangle$ and $|\mathcal{E}_{K'}\rangle$,

$$\begin{aligned} |X_K\rangle |\mathcal{E}_0\rangle &\rightarrow |X_K\rangle |\mathcal{E}_K\rangle, \\ |X_{K'}\rangle |\mathcal{E}_0\rangle &\rightarrow |X_{K'}\rangle |\mathcal{E}_{K'}\rangle, \end{aligned} \quad (45)$$

the biexciton off-diagonal coherence acquires a suppression factor $\langle \mathcal{E}_K | \mathcal{E}_{K'} \rangle$, which sets the phase-scan visibility

(see Appendix D 1). The extent of this leakage depends on how long the intermediate exciton is exposed to the phonon bath before the second photon arrives. The sinc-shaped JSA of type-II SPDC has a total temporal correlation window $[-T_e/2, T_e/2]$; causality (the second photon must arrive after the first) restricts the intermediate-state dwell time to the positive half $[0, T_e/2]$ (Appendix B). The exciton dephasing further down-weights long dwell times. The net result is a mean intermediate-exciton dwell time (Appendix D 1)

$$\langle \tau \rangle = \frac{1}{\gamma_X} - \frac{T_e/2}{e^{\gamma_X T_e/2} - 1}, \quad (46)$$

For broadband SPDC ($T_e \sim 100$ fs) with $\gamma_X \sim 1.5$ meV, $\gamma_X T_e \approx 0.23 \ll 1$, placing this regime in the broad-bandwidth limit $\langle \tau \rangle \rightarrow T_e/4$ of Eq. (46), giving $\langle \tau \rangle \approx 25$ fs, one to two orders of magnitude shorter than τ_{valley} , ensuring that the intermediate exciton has minimal exposure to valley decoherence. Even in the narrow-bandwidth (CW) limit, $\langle \tau \rangle$ saturates at $1/\gamma_X$, remaining short on the valley-decoherence timescale.

B. Valley dephasing and intervalley scattering

Valley dephasing and intervalley scattering contribute multiplicatively to the total visibility. Valley dephasing accumulates which-valley information at rate γ_{valley} during $\langle \tau \rangle$. The valley depolarization time τ_v measured by Kerr rotation ($\tau_v \approx 6$ ps for the bright *A*-exciton in WSe₂ at 4 K [53]) characterizes population relaxation between valleys, driven primarily by the Coulomb exchange (Maialle-Silva-Sham) mechanism [54, 55]. The valley coherence time τ_{vc} , which governs the decay of intervalley superpositions, is generally shorter, with $1/\tau_{vc} = 1/(2\tau_v) + \gamma_v^*$, where γ_v^* is the pure valley dephasing rate [24, 54]. The quantity entering the visibility formula is the valley decoherence rate $\gamma_{\text{valley}} \equiv 1/\tau_{vc}$, i.e., the rate at which the phonon environment acquires which-valley information. Averaging the decoherence factor over the sinc-model intermediate-state time distribution gives (Appendix D 1)

$$V_{\text{deph}} = \frac{\gamma_X}{\gamma_X + \gamma_{\text{valley}}} \frac{1 - e^{-(\gamma_X + \gamma_{\text{valley}})T_e/2}}{1 - e^{-\gamma_X T_e/2}}. \quad (47)$$

With $\hbar\gamma_X \approx 1.5$ meV [46, 47, 56] and valley coherence times $\tau_{vc} \sim 1$ –3 ps ($\hbar\gamma_{\text{valley}} \approx 0.2$ –0.7 meV, as inferred from linearly polarized photoluminescence experiments [57, 58]), broadband SPDC ($T_e = 100$ fs) gives $V_{\text{deph}} \approx 0.97$; moderate bandwidth ($T_e = 500$ fs) gives $V_{\text{deph}} \approx 0.89$; even the CW limit saturates at $V_{\text{deph}} \rightarrow \gamma_X/(\gamma_X + \gamma_{\text{valley}}) \approx 0.70$.

The Markov ansatz [Eq. (D3), Appendix D 1] underpinning these estimates assumes an exponential decay of the environmental overlap, which is valid when the exciton dwell time exceeds the bath correlation time $\tau_c \sim$

tens of femtoseconds set by the acoustic-phonon bandwidth. At shorter dwell times the true decoherence onset is slower than exponential, so the Markov-based V_{deph} values are conservative lower bounds in the broadband regime ($T_e \lesssim 100$ fs, $\langle \tau \rangle \approx 25$ fs); for $T_e \gtrsim 2\tau_c$ the exponential form is recovered and the estimates are quantitative.

Intervalley scattering physically transfers an exciton between valleys, corrupting the pathway mapping. In monolayer WSe₂, this requires large momentum transfer and is mediated by short-range disorder or zone-edge phonons, placing τ_{iv} at or above the measured valley depolarization time $\tau_v \approx 6$ ps at 4 K [53]. Averaging the survival probability $e^{-\gamma_{iv}\tau}$, where τ is the intermediate-state dwell time, over the same dwell-time distribution used for V_{deph} (see Appendix D 2) gives the scattering probability

$$P_{iv} = 1 - \frac{\gamma_X}{\gamma_X + \gamma_{iv}} \frac{1 - e^{-(\gamma_X + \gamma_{iv})T_e/2}}{1 - e^{-\gamma_X T_e/2}}, \quad (48)$$

giving $P_{iv} \approx 0.2\%$ for broadband SPDC with $\tau_{iv} = 10$ ps, a representative value consistent with the lower bound above. The combined material-side visibility is

$$V_{\text{mat}} = V_{\text{deph}} \times (1 - P_{iv}). \quad (49)$$

The observable emission intensity is then $I(\varphi) \propto 1 + V_{\text{mat}} \cos \varphi$.

Fig. 5(a) shows V_{mat} (left axis) and P_{iv} (right axis) as functions of T_e . In the broadband regime ($T_e \lesssim 100$ fs), all three τ_{vc} curves converge near $V_{\text{mat}} \approx 0.97$ –0.99 because the intermediate-exciton dwell time is so short ($\langle \tau \rangle \lesssim 25$ fs) that valley decoherence barely acts regardless of sample quality. As T_e increases, the curves fan out because the exciton spends progressively more time exposed to the phonon bath, and samples with shorter valley coherence ($\tau_{vc} = 1$ ps) degrade faster than cleaner samples ($\tau_{vc} = 3$ ps). In the CW limit ($T_e \rightarrow \infty$), each curve saturates at $V_{\text{deph}} \rightarrow \gamma_X/(\gamma_X + \gamma_{\text{valley}})$, giving ≈ 0.70 , 0.82, and 0.87 for $\tau_{vc} = 1, 2$, and 3 ps, respectively. The intervalley scattering probability P_{iv} (green curve) is independent of τ_{vc} and remains below $\sim 4\%$ across the entire T_e range, confirming that it is a minor correction to the visibility budget.

C. Dark-to-bright relaxation within the triplet pair

Under triplet-tuned excitation, the dark eigenstate Φ_3 is not primarily threatened by relaxation to Φ_1 (which is ~ 7.5 meV lower in energy); the relevant nearby bright channel is the exchange-split partner Φ_2 , which lies only about 0.7 meV above Φ_3 as shown in Table I. This splitting is comparable to the homogeneous exciton linewidth $\hbar\gamma_X \approx 1.5$ meV [46, 47, 56], so under broadband excitation the pump addresses both eigenstates nearly equally. Even a narrow-bandwidth source achieves only partial spectral discrimination, since the Lorentzian absorption

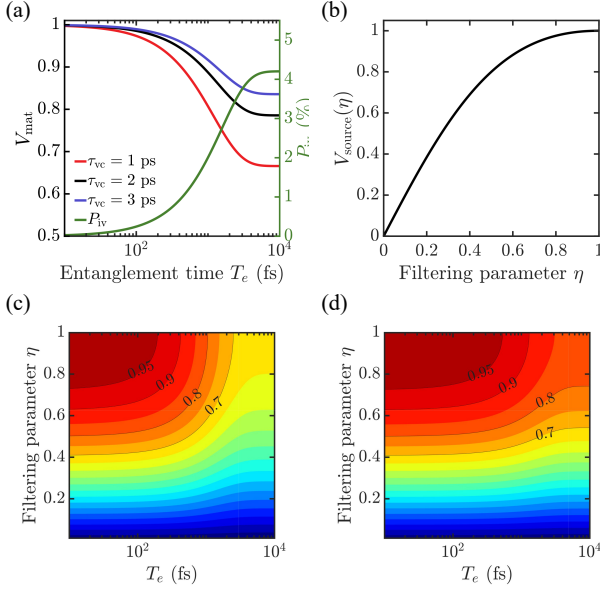


FIG. 5. Decoherence and source-imperfection limits on the eigenstate selection rule. (a) Material-side visibility V_{mat} (left axis) as a function of entanglement time T_e for valley coherence times $\tau_{vc} = 1, 2$, and 3 ps at 4 K. The intervalley scattering probability P_{iv} (right axis, green) remains below $\sim 5\%$ across the full range. Broadband SPDC ($T_e \sim 100$ fs) yields $V_{\text{mat}} \gtrsim 0.97$. (b) Source-side visibility $V_{\text{source}}(\eta)$ [Eq. (54)] as a function of the polarization-filtering parameter η , quantifying the effect of polarization-dependent loss on the phase-scan contrast. (c), (d) Total visibility $V_{\text{total}} = V_{\text{mat}} \times V_{\text{source}}$ as a function of both T_e and η for $\tau_{vc} = 1$ ps (c) and 3 ps (d). Contour lines mark $V_{\text{total}} = 0.70, 0.80, 0.90$, and 0.95 .

tails of Φ_2 and Φ_3 overlap, as shown in Figs. 3(b) and 4(b). Consequently, the $\Phi_3 \rightarrow \Phi_2$ relaxation product cannot be removed by spectral filtering and contributes directly to the bright-sector background. This makes the $\Phi_3 \leftrightarrow \Phi_2$ relaxation rate a principal kinetic unknown for the present scheme. The exchange model fixes the eigenvectors and splittings but does not determine this rate; reported biexciton lifetimes in monolayer TMDs lie in the picosecond regime [24, 29, 59], yet the lifetime of the individual exchange-dark eigenstate Φ_3 has not been measured directly.

The exchange-symmetry selection rule offers a route to such a measurement. At $\varphi = \pi$, the Bell state populates Φ_3 exactly while the Φ_2 population vanishes by the geometric overlap, not by spectral selectivity [Eq. (37)]; the classical state of Eq. (42) similarly suppresses Φ_2 at $\theta = \pi$, but at half the Φ_3 rate (Table II), which is the upper bound for any classical source [Eq. (41)]. This symmetry-based initial-state preparation could serve as a direct probe of the $\Phi_3 \rightarrow \Phi_2$ inter-eigenstate relaxation rate. For the symmetry-based extinction to remain experimentally observable, this rate must satisfy

$$\Gamma_{\Phi_3 \rightarrow \Phi_2} \ll \Gamma_{\text{loss}}^{(\Phi_3)}, \quad (50)$$

where $\Gamma_{\text{loss}}^{(\Phi_3)}$ denotes the total rate for nonradiative decay and any leakage out of the biexciton detection channel; that is, dark-to-bright transfer within the triplet must remain slower than loss from the prepared dark state. Reported biexciton lifetimes in monolayer TMDs lie in the picosecond regime [24, 29, 59], but whether this timescale applies to the exchange-dark eigenstate Φ_3 specifically has not been established experimentally, which is why the time-resolved cascade emission at $\varphi = \pi$ can itself serve as the measurement.

If a fraction of the Φ_3 population transfers to the bright sector before decay, the leading effect is a phase-independent background added to the bright-sector intensity, reducing the visibility without generating a spurious $\cos \varphi$ modulation. The material-side visibility reported in Sec. IIIB and shown in Fig. 5(a) is therefore an upper bound that assumes negligible transfer; any finite transfer reduces it while leaving the selection-rule structure intact. When Eq. (50) fails, the time-resolved cascade emission at $\varphi = \pi$ exhibits a delayed rise with time constant $\Gamma_{\Phi_3 \rightarrow \Phi_2}^{-1}$ relative to the prompt Φ_2 emission at $\varphi = 0$, separating the kinetic contribution from the symmetry-based extinction.

D. Polarization-dependent loss and partial entanglement

Polarization-dependent losses in the excitation path (stress birefringence in optical elements, imperfect waveplate alignment, or unbalanced mirror reflectivities) attenuate one circular component relative to the other. We model this as a local filter $F = \text{diag}(1, \eta)$ acting on the signal mode ($0 < \eta \leq 1$), which transforms the Bell state $|\psi(\varphi)\rangle_{\text{Bell}}$ [Eq. (30)] into

$$|\psi'\rangle \propto |\sigma^+\rangle_s |\sigma^-\rangle_i + \eta e^{i\varphi} |\sigma^-\rangle_s |\sigma^+\rangle_i, \quad (51)$$

since F passes $|\sigma^+\rangle$ unchanged and attenuates $|\sigma^-\rangle$ by η . After normalization, the polarization density matrix elements become

$$p_{\sigma^+\sigma^-} = \frac{1}{1+\eta^2}, \quad p_{\sigma^-\sigma^+} = \frac{\eta^2}{1+\eta^2}, \quad (52)$$

$$\rho_{\sigma^+\sigma^-, \sigma^-\sigma^+} = \frac{\eta e^{-i\varphi}}{1+\eta^2}.$$

Substituting into Eq. (32) and using the exchange symmetry $\langle \Phi_\alpha | B_2 \rangle = \langle \Phi_\alpha | B_3 \rangle \equiv g_\alpha$ for the bright eigenstates gives

$$W_\alpha(\varphi) = \kappa_\alpha |g_\alpha|^2 [p_{\sigma^+\sigma^-} + p_{\sigma^-\sigma^+} + 2\text{Re}(\rho_{\sigma^+\sigma^-, \sigma^-\sigma^+})]$$

$$= \kappa_\alpha |g_\alpha|^2 \left[1 + \frac{2\eta}{1+\eta^2} \cos \varphi \right]. \quad (53)$$

Reading off the coefficient of $\cos \varphi$ yields the visibility

$$V_{\text{source}}(\eta) = \frac{2\eta}{1+\eta^2} = \text{sech}(\gamma), \quad \eta = e^{-\gamma}, \quad (54)$$

which interpolates smoothly from unity (perfect source, $\eta = 1$) to zero (complete depolarization, $\eta \rightarrow 0$), as shown in Fig. 5(b). The concurrence C is a standard measure of two-qubit entanglement, ranging from 0 for a separable state to 1 for a maximally entangled Bell state [60]. Under the local filtering $F = \text{diag}(1, \eta)$, the concurrence of the filtered state reduces to $C'(\eta) = 2\eta/(1 + \eta^2)$ following the framework of Verstraete et al. [61], which is identical to $V_{\text{source}}(\eta)$ [Eq. (54)]. This identity holds specifically for an initial pure Bell state under local filtering on one arm; extensions to mixed or non-Bell biphoton sources would require the full Wootters formula [60] to evaluate the residual concurrence. Within the local-filtering model considered here, the phase-scan visibility therefore tracks the residual entanglement of the source, so that degradation of the biphoton concurrence is reflected as a reduction in the eigenstate-selection contrast.

The full experimental visibility, combining the material-side [Eq. (49)] and source-side [Eq. (54)] factors, is

$$V_{\text{total}} = V_{\text{mat}} \times V_{\text{source}}(\eta). \quad (55)$$

Figures 5(c) and 5(d) map V_{total} over the full (T_e, η) plane for $\tau_{vc} = 1$ and 3 ps, respectively. For broadband SPDC ($T_e \sim 100$ fs) with high source quality ($\eta \gtrsim 0.95$), all three τ_{vc} curves yield $V_{\text{total}} \gtrsim 0.97$, and cleaner samples ($\tau_{vc} = 3$ ps) maintain $V_{\text{total}} \gtrsim 0.95$ out to $T_e \lesssim 500$ fs, relaxing the bandwidth requirements on the SPDC source.

IV. DISCUSSION AND CONCLUSION

We have shown that Bell-family photon pairs map their phase φ onto the symmetric/antisymmetric biexciton sectors in monolayer WSe₂, with $\varphi = 0$ driving the bright eigenstates and $\varphi = \pi$ driving the exchange-dark eigenstate Φ_3 . The Bell state's dark-eigenstate pumping rate is at least twice that of any classical source satisfying the PPT bound Eq. (38), because the cross-circular Bell state confines all flux to the cross-circular sector—a spectral fingerprint that no separable polarization state reproduces. The extinction at $\varphi = \pi$ is enforced by the $B_2 \leftrightarrow B_3$ exchange symmetry and holds at any pump detuning [Eq. (37)], and is relatively robust against phonon decoherence ($V \gtrsim 0.97$ for broadband SPDC at 4 K with high source purity). Furthermore, within the local-filtering model, the phase-scan visibility tracks the source concurrence ($V_{\text{source}} = C'(\eta)$ for an initial Bell state), suggesting that the biexciton manifold could serve as a solid-state readout of the source concurrence. To our knowledge, this operational identity between phase-scan visibility and two-qubit concurrence has not been previously articulated in the ETPA literature. The ongoing effort to measure ETPA cross sections in molecular systems [8–11] targets a quantitative rate enhancement that must be separated from single-photon backgrounds. The eigenstate selection rule proposed here

offers a structurally different approach in which the entanglement signature is a symmetry-enforced redistribution of biexciton population, observable as the presence or absence of cascade emission in a φ scan, rather than an absolute rate comparison.

The numerical binding energies and exchange splittings in Table I are computed for WSe₂ on sapphire [33]. The eigenvector structure (which eigenstates are symmetric and antisymmetric under $B_2 \leftrightarrow B_3$) is determined to leading order by the crystal symmetry of the TMD monolayer rather than the substrate, so the selection-rule zeros are substrate-insensitive and only the quantitative binding energies and visibility estimates should shift under hBN encapsulation. The exciton linewidth used in this work ($\hbar\gamma_X \approx 1.5$ meV) is taken from hBN-encapsulated samples [46, 47], ensuring internal consistency. The selection rule relies on valley symmetry ($T_2(\omega) = T_3(\omega)$), which is guaranteed by time-reversal symmetry at zero magnetic field. In hBN-encapsulated WSe₂ this condition is well satisfied, with reported valley dipole asymmetries well below 10% [46, 47]. The effects of explicit symmetry breaking by applied magnetic fields or strong uniaxial strain are beyond the scope of the present work.

Finally, the symmetry argument is not unique to WSe₂. The required ingredients are (i) two degenerate absorption pathways that create the same final state in different internal configurations, (ii) an exchange interaction that splits the configuration space into symmetric and antisymmetric eigenstates, and (iii) a precharacterized photon source whose relative phase controls those pathways. Other TMD monolayers (MoSe₂, WS₂, MoS₂) [29–32, 37, 38] share the same valley-based symmetry logic and are natural candidates, since the eigenvector structure (symmetric vs. antisymmetric under $B_2 \leftrightarrow B_3$) is dictated by the C_{3h} point group common to all group-VI TMD monolayers, with only the numerical exchange splittings varying between materials.

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Appendix A: Two-Photon Transition Amplitude

Inserting a complete set of intermediate one-exciton states $\sum_n |n\rangle \langle n| = \mathbb{I}$ between the two interactions in Eq. (5), the interaction-picture time dependence generates transition-frequency phases:

$$\begin{aligned} & \langle B_j | \hat{d}_{\sigma_2}(t_2) \hat{d}_{\sigma_1}(t_1) | g \rangle \\ &= \sum_n \langle B_j | \hat{d}_{\sigma_2} | n \rangle \langle n | \hat{d}_{\sigma_1} | g \rangle e^{i\omega_{B_j n} t_2} e^{i\omega_{n g} t_1}, \end{aligned} \quad (A1)$$

where $\omega_{B_j n} = (E_{B_j} - E_n)/\hbar$ and $\omega_{ng} = (E_n - E_g)/\hbar$. After the first interaction at t_1 , the system spends time $\tau = t_2 - t_1$ in the intermediate state $|n\rangle$, during which the optical coherence ρ_{ng} dephases at rate γ_n . After the second interaction at t_2 , the system spends time $\tau' = t - t_2$ in the final state $|B_j\rangle$, during which $\rho_{B_j g}$ dephases at rate γ_B . Including these causal decay factors:

$$e^{i\omega_{B_j n} t_2} e^{i\omega_{ng} t_1} \rightarrow e^{i\omega_{B_j n} t_2} e^{i\omega_{ng} t_1} e^{-\gamma_n(t_2-t_1)} e^{-\gamma_B(t-t_2)}. \quad (\text{A2})$$

In the interaction picture, each positive-frequency field operator carries a phase $e^{-i\omega t}$ at the time of the interaction, so the signal photon annihilated at t_2 contributes $e^{-i\omega_s t_2}$ and the idler photon annihilated at t_1 contributes $e^{-i\omega_i t_1}$. Combining these field phases with the matrix element Eq. (10) for Contraction A ($\hat{a}_s$ at t_2 , $\hat{a}_i$ at t_1), the matter response Eq. (A2), and the $(-i/\hbar)^2$ prefactor from Eq. (5):

$$\begin{aligned} \mathcal{M}_j^{(A)}(t) &= \frac{-1}{\hbar^2} \sum_n \iint d\omega_s d\omega_i \Psi(\omega_s, \sigma_2; \omega_i, \sigma_1) \\ &\times \langle B_j | \hat{d}_{\sigma_2} | n \rangle \langle n | \hat{d}_{\sigma_1} | g \rangle \int_{-\infty}^t dt_2 \int_{-\infty}^{t_2} dt_1 \\ &\times e^{-\gamma_B(t-t_2)} e^{i\omega_{B_j n} t_2} e^{i\omega_{ng} t_1} \\ &\times e^{-\gamma_n(t_2-t_1)} e^{-i\omega_s t_2} e^{-i\omega_i t_1}. \end{aligned} \quad (\text{A3})$$

Switching to the dwell-time variables τ and τ' introduced above as integration variables,

$$t_2 = t - \tau', \quad t_1 = t - \tau' - \tau, \quad \tau, \tau' \geq 0,$$

the Jacobian is unity and both integration limits become $[0, \infty)$.

The final-state damping ($t_2 \rightarrow t$) becomes

$$e^{-\gamma_B(t-t_2)} = e^{-\gamma_B \tau'}.$$

The intermediate-state phases give

$$\begin{aligned} e^{i\omega_{B_j n} t_2} e^{i\omega_{ng} t_1} &= e^{i(\omega_{B_j n} + \omega_{ng})t} e^{-i(\omega_{B_j n} + \omega_{ng})\tau'} e^{-i\omega_{ng}\tau} \\ &= e^{i\omega_{B_j} t} e^{-i\omega_{B_j} \tau'} e^{-i\omega_{ng}\tau}, \end{aligned}$$

where we used $\omega_{B_j n} + \omega_{ng} = \omega_{B_j g} = \omega_{B_j}$, having set $E_g = 0$ as the energy reference. The field phases (Contraction A, signal at t_2 , idler at t_1) become

$$e^{-i\omega_s t_2} e^{-i\omega_i t_1} = e^{-i(\omega_s + \omega_i)t} e^{i(\omega_s + \omega_i)\tau'} e^{i\omega_i \tau}.$$

Collecting by variable, the t -dependent overall phase $e^{i(\omega_{B_j} - \omega_s - \omega_i)t}$ factors out, and the remaining integrand

separates. In the new variables, Eq. (A3) becomes

$$\begin{aligned} \mathcal{M}_j^{(A)}(t) &= -\frac{1}{\hbar^2} \sum_n \iint d\omega_s d\omega_i \Psi(\omega_s, \sigma_2; \omega_i, \sigma_1) \\ &\times \langle B_j | \hat{d}_{\sigma_2} | n \rangle \langle n | \hat{d}_{\sigma_1} | g \rangle e^{i(\omega_{B_j} - \omega_s - \omega_i)t} \\ &\times \int_0^\infty d\tau' e^{[i(\omega_s + \omega_i - \omega_{B_j}) - \gamma_B]\tau'} \\ &\times \int_0^\infty d\tau e^{[-i(\omega_{ng} - \omega_i) - \gamma_n]\tau}. \end{aligned} \quad (\text{A4})$$

Both integrals are over positive time differences with exponential damping, so both converge automatically ($\gamma_n, \gamma_B > 0$).

Final-state integral (τ'):

$$\begin{aligned} \int_0^\infty d\tau' e^{[i(\omega_s + \omega_i - \omega_{B_j}) - \gamma_B]\tau'} &= \frac{1}{\gamma_B - i(\omega_s + \omega_i - \omega_{B_j})} \\ &= \frac{-i}{(\omega_{B_j} - \omega_s - \omega_i) - i\gamma_B} \equiv L(\omega_s + \omega_i). \end{aligned} \quad (\text{A5})$$

This is the final-state Lorentzian enforcing two-photon energy conservation.

Intermediate-state integral (τ):

$$\begin{aligned} \int_0^\infty d\tau e^{[-i(\omega_{ng} - \omega_i) - \gamma_n]\tau} &= \frac{1}{\gamma_n + i(\omega_{ng} - \omega_i)} \\ &= \frac{-i}{(\omega_{ng} - \omega_i) - i\gamma_n}, \end{aligned} \quad (\text{A6})$$

which is the retarded intermediate-state propagator. In Contraction A the photon absorbed at t_1 has frequency ω_i , so the intermediate-state propagator is evaluated at ω_i .

Combining all factors, Contraction A gives:

$$\begin{aligned} \mathcal{M}_j^{(A)} &\propto \iint d\omega_s d\omega_i \Psi(\omega_s, \sigma_2; \omega_i, \sigma_1) \\ &\times \sum_n \frac{\langle B_j | \hat{d}_{\sigma_2} | n \rangle \langle n | \hat{d}_{\sigma_1} | g \rangle}{[\hbar(\omega_{ng} - \omega_i) - i\hbar\gamma_n]} L(\omega_s + \omega_i). \end{aligned} \quad (\text{A7a})$$

For Contraction B (idler at t_2 , signal at t_1), the JSA component is $\Psi(\omega_s, \sigma_1; \omega_i, \sigma_2)$ [Eq. (11)], indices swapped and the field phases are $e^{-i\omega_i t_2} e^{-i\omega_s t_1}$. The only change is in the τ -integral, where the photon absorbed at t_1 now has frequency ω_s instead of ω_i :

$$\int_0^\infty d\tau e^{[-i(\omega_{ng} - \omega_s) - \gamma_n]\tau} = \frac{-i}{(\omega_{ng} - \omega_s) - i\gamma_n}.$$

The τ' -integral is unchanged (it depends only on the total photon energy $\omega_s + \omega_i$). Thus:

$$\begin{aligned} \mathcal{M}_j^{(B)} &\propto \iint d\omega_s d\omega_i \Psi(\omega_s, \sigma_1; \omega_i, \sigma_2) \\ &\times \sum_n \frac{\langle B_j | \hat{d}_{\sigma_2} | n \rangle \langle n | \hat{d}_{\sigma_1} | g \rangle}{[\hbar(\omega_{ng} - \omega_s) - i\hbar\gamma_n]} L(\omega_s + \omega_i). \end{aligned} \quad (\text{A7b})$$

Adding both contractions, the total amplitude $\mathcal{M}_j = \mathcal{M}_j^{(A)} + \mathcal{M}_j^{(B)}$ shares the common final-state Lorentzian $L(\omega_s + \omega_i)$. Defining the frequency-dependent two-photon matrix element $T_j(\omega)$ [Eq. (12)] absorbs the $\hbar$ prefactors and the sum over intermediate states, yielding the total amplitude [Eq. (13)].

The non-degenerate excitation scheme places the signal at the exciton resonance ($\omega_s = E_X/\hbar$) and the idler off-resonance by the biexciton binding energy ($\omega_i = \omega_s - E_B/\hbar$). The frequency-dependent kernel $T(\omega)$ is a Lorentzian centered on ω_X :

$$T(\omega) = \frac{d_0 d_{XX,X}}{\hbar(\omega_X - \omega) - i\hbar\gamma_X}, \quad (\text{A8})$$

where $d_0 = \langle X_K | \hat{d}_+ | g \rangle$ is the exciton oscillator strength (Eq. (8)) and $d_{XX,X} = \langle B_2 | \hat{d}_- | X_K \rangle$ is the exciton-to-biexciton transition dipole. The squared modulus of $T(\omega)$ at the two photon frequencies reproduces the suppression ratio ≈ 0.006 stated in Eq. (19).

Appendix B: Frequency Integral for Type-II SPDC

The dominant-contraction amplitude from the main text (Sec. II B 3), where valley symmetry gives $T_2(\omega_s) = T_3(\omega_s) \equiv T(\omega_s)$:

$$\mathcal{M}_2 \approx \frac{1}{\sqrt{2}} \iint d\omega_s d\omega_i f(\omega_s, \omega_i) T(\omega_s) L(\omega_s + \omega_i).$$

For a CW (monochromatic) pump at $\omega_p = \omega_s + \omega_i$, which is valid when the pump linewidth is much narrower than the biexciton linewidth γ_B (a condition readily satisfied by the single-mode CW lasers used to pump SPDC sources), the JSA takes the standard form [4]:

$$f(\omega_s, \omega_i) = \mathcal{N} \delta(\omega_s + \omega_i - \omega_p) \text{sinc}[(\omega_X - \omega_s)T_e/2], \quad (\text{B1})$$

where $\text{sinc}[(\omega_X - \omega_s)T_e/2]$ is the marginal signal spectral amplitude, T_e is the entanglement time set by the crystal length and group-velocity mismatch, and $\mathcal{N}$ is a normalization constant. Substituting into the amplitude:

$$\mathcal{M}_2 = \frac{\mathcal{N}}{\sqrt{2}} \iint d\omega_s d\omega_i \delta(\omega_s + \omega_i - \omega_p) \times \text{sinc}[(\omega_X - \omega_s)T_e/2] T(\omega_s) L(\omega_s + \omega_i).$$

The δ -function eliminates the ω_i integration, setting $\omega_i = \omega_p - \omega_s$ and hence $\omega_s + \omega_i = \omega_p$. The final-state factor becomes a constant $L(\omega_p)$:

$$\mathcal{M}_2 = \frac{\mathcal{N}L(\omega_p)}{\sqrt{2}} \int_{-\infty}^{\infty} d\omega_s \text{sinc}[(\omega_X - \omega_s)T_e/2] T(\omega_s).$$

We change the integration variable from ω_s to the signal detuning from the intermediate exciton, $\Delta \equiv \omega_X - \omega_s$:

$$\mathcal{M}_2 = \frac{\mathcal{N}L(\omega_p)}{\sqrt{2}} \int_{-\infty}^{\infty} d\Delta \text{sinc}(\Delta T_e/2) T(\omega_X - \Delta).$$

The valley selection rules restrict the intermediate-state sum in $T(\omega)$ [Eq. (12)] to a single exciton $|X_K\rangle$ at frequency ω_X , giving

$$\begin{aligned} T(\omega_X - \Delta) &= \frac{d_0 d_{XX,X}}{\hbar(\omega_X - (\omega_X - \Delta)) - i\hbar\gamma_X} \\ &= \frac{d_0 d_{XX,X}}{\hbar(\Delta - i\gamma_X)}, \end{aligned}$$

Substituting:

$$\mathcal{M}_2 = \frac{\mathcal{N}L(\omega_p) d_0 d_{XX,X}}{\sqrt{2}\hbar} \int_{-\infty}^{\infty} d\Delta \frac{\text{sinc}(\Delta T_e/2)}{\Delta - i\gamma_X}. \quad (\text{B2})$$

The source-dependent prefactor $\mathcal{N}L(\omega_p)/\sqrt{2}$ is common to all pathways and will be absorbed into the rate prefactor κ_α below. The pathway-specific physics is encoded entirely in the spectral integral $\int d\Delta \text{sinc}(\Delta T_e/2)/(\Delta - i\gamma_X)$, which we now evaluate.

For the physical sinc spectral amplitude, we represent the sinc as a Fourier integral over a rectangular temporal window:

$$\text{sinc}\left(\frac{\Delta T_e}{2}\right) = \frac{1}{T_e} \int_{-T_e/2}^{T_e/2} e^{i\Delta t} dt.$$

Substituting into the spectral integral Eq. (B2) and exchanging the order of integration, the Δ -integral evaluates by contour integration. The sole pole at $\Delta = i\gamma_X$ lies in the upper half-plane, while the convergence factor $e^{i\Delta t}$ decays in the upper (lower) half-plane for $t > 0$ ($t < 0$). Closing in the upper half-plane for $t > 0$ picks up the pole; closing in the lower half-plane for $t < 0$ encloses no poles:

$$\int_{-\infty}^{\infty} \frac{e^{i\Delta t}}{\Delta - i\gamma_X} d\Delta = \begin{cases} 2\pi i e^{-\gamma_X t}, & t > 0, \\ 0, & t < 0. \end{cases}$$

Only the $t > 0$ portion of the rectangular window $[-T_e/2, T_e/2]$ contributes, leaving an elementary integral:

$$\int_0^{T_e/2} e^{-\gamma_X t} dt = \frac{1}{\gamma_X} (1 - e^{-\gamma_X T_e/2}).$$

Collecting the prefactor $2\pi i/T_e$ from the sinc representation, the spectral integral evaluates to

$$\int_{-\infty}^{\infty} d\Delta \frac{\text{sinc}(\Delta T_e/2)}{\Delta - i\gamma_X} = \frac{2\pi i}{T_e \gamma_X} (1 - e^{-\gamma_X T_e/2}). \quad (\text{B3})$$

Substituting Eq. (B3) into Eq. (B2) gives the full amplitude $\mathcal{M}_2$, from which we identify the effective matrix element by grouping the source-dependent factors $\mathcal{N}L(\omega_p)(2\pi/T_e)$ into the prefactor that will become κ_α :

$$\mathcal{M}_2 = \frac{2\pi \mathcal{N}L(\omega_p)}{\sqrt{2}T_e} \frac{d_0 d_{XX,X}}{\hbar} \frac{i}{\gamma_X} (1 - e^{-\gamma_X T_e/2}).$$

The resonant two-photon amplitude is therefore

$$\mathcal{T} = \frac{d_0 d_{XX,X}}{\hbar} \frac{i}{\gamma_X} (1 - e^{-\gamma_X T_e/2}), \quad (\text{B4})$$

which is Eq. (26) of the main text; the narrow- and broad-bandwidth limits are discussed there.

The $|B_3\rangle$ pathway proceeds through $|X_{K'}\rangle$ instead of $|X_K\rangle$, with $\sigma^+ \leftrightarrow \sigma^-$ polarizations. By time-reversal symmetry at zero magnetic field, all dipole matrix elements and dephasing rates are identical under $K \leftrightarrow K'$, yielding

$$\mathcal{T}(\sigma^-\sigma^+) = \mathcal{T}(\sigma^+\sigma^-) = \mathcal{T}. \quad (\text{B5})$$

Appendix C: Eigenstate-Resolved Pumping Rate

We now derive the resonance-weighted pumping-rate formula Eq. (27) by projecting the Dyson-series amplitudes onto the eigenstate basis.

The Dyson-series amplitudes derived in Appendices A and B give the transition amplitude from the ground state to each biexciton configuration. For a pure biphoton polarization state $|\psi_{2\text{ph}}\rangle_{\text{pol}} = \sum_{\mu_1\mu_2} c_{\mu_1\mu_2} |\mu_1\mu_2\rangle$, each polarization pair (μ_1, μ_2) activates exactly one configuration $B_{\mu_1\mu_2}$ [Eqs. (9a) and (9b)] with amplitude proportional to $c_{\mu_1\mu_2} \mathcal{T}$ [Eq. (B5)]. The configuration-basis

superposition imprinted by the photon pair is therefore

$$|\psi_{\text{prep}}\rangle \propto \sum_{\mu_1\mu_2} c_{\mu_1\mu_2} |B_{\mu_1\mu_2}\rangle, \quad (\text{C1})$$

where $\mathcal{T}$ and all source-dependent spectral factors have been suppressed temporarily.

The transition amplitude from the ground state to eigenstate $|\Phi_\alpha\rangle$, denoted $\widetilde{\mathcal{M}}_\alpha$, is obtained by projecting the configuration-basis state onto $\langle\Phi_\alpha| = \sum_j \langle\Phi_\alpha|B_j\rangle \langle B_j|$ (Table I) and restoring the suppressed spectral prefactors. In doing so, the final-state Lorentzian $L(\omega_s + \omega_i)$ from the configuration-basis derivation [Eq. (A5)] must be evaluated at the eigenstate energy $\omega_{\Phi_\alpha g}$ rather than the configuration energy $\omega_{B_j g}$, because the true energy eigenstates of the biexciton manifold are the exchange-split states $|\Phi_\alpha\rangle$. This gives

$$\widetilde{\mathcal{M}}_\alpha = \frac{2\pi \mathcal{N}}{\sqrt{2} T_e} L_\alpha(\omega_p) \mathcal{T} \sum_{\mu_1\mu_2} c_{\mu_1\mu_2} \langle\Phi_\alpha|B_{\mu_1\mu_2}\rangle, \quad (\text{C2})$$

with $L_\alpha(\omega_p) = [(\omega_{\Phi_\alpha g} - \omega_p) - i\gamma_B]^{-1}$. The transition probability per incident pair is therefore

$$\begin{aligned} P_\alpha &= |\widetilde{\mathcal{M}}_\alpha|^2 = \frac{2\pi^2 |\mathcal{N}|^2}{T_e^2} |L_\alpha(\omega_p)|^2 |\mathcal{T}|^2 \left| \sum_{\mu_1\mu_2} c_{\mu_1\mu_2} \langle\Phi_\alpha|B_{\mu_1\mu_2}\rangle \right|^2 \\ &= \frac{2\pi^2 |\mathcal{N}|^2}{T_e^2} |L_\alpha(\omega_p)|^2 |\mathcal{T}|^2 \sum_{\mu_1\mu_2, \mu'_1\mu'_2} \langle\Phi_\alpha|B_{\mu_1\mu_2}\rangle \langle B_{\mu'_1\mu'_2}|\Phi_\alpha\rangle c_{\mu_1\mu_2} c_{\mu'_1\mu'_2}^*. \end{aligned} \quad (\text{C3})$$

For a mixed biphoton state $\rho = \sum_k p_k |\psi_k\rangle \langle\psi_k|$, the ensemble average replaces $c_{\mu_1\mu_2} c_{\mu'_1\mu'_2}^*$ by the polarization density matrix $\rho_{\mu_1\mu_2, \mu'_1\mu'_2}$. Using the linear-in-flux scaling of ETPA in the isolated-pair regime [3, 16], the pumping rate is

$$W_\alpha = \kappa_\alpha \sum_{\mu_1\mu_2, \mu'_1\mu'_2} \langle\Phi_\alpha|B_{\mu_1\mu_2}\rangle \langle B_{\mu'_1\mu'_2}|\Phi_\alpha\rangle \rho_{\mu_1\mu_2, \mu'_1\mu'_2}, \quad (\text{C4})$$

with

$$\kappa_\alpha \equiv \kappa_0 \frac{\gamma_B^2}{(\omega_{\Phi_\alpha g} - \omega_p)^2 + \gamma_B^2}, \quad \kappa_0 \equiv R_{\text{pairs}} \frac{2\pi^2 |\mathcal{N}|^2 |\mathcal{T}|^2}{T_e^2 \gamma_B^2}.$$

The factorization now separates the eigenstate geometry and light correlations exactly as before, but it makes clear that the final-state resonance is not common to all biexciton eigenstates. In the equal- L_α limit one recovers the equal-resonance approximation $\kappa_\alpha \rightarrow \kappa_0$.

Appendix D: Coherence Preservation and Visibility Estimates

1. Tripartite state, decoherence factor, and mean dwell time

After the signal photon is absorbed from $|\psi(\varphi)\rangle_{\text{Bell}}$ [Eq. (30)], but before the idler arrives, the joint state of (material + idler photon + environment) is

$$|\psi_{\text{tot}}(t)\rangle = \frac{1}{\sqrt{2}} \left(|X_K\rangle |\sigma^-\rangle_i |\mathcal{E}_K(t)\rangle + e^{i\varphi} |X_{K'}\rangle |\sigma^+\rangle_i |\mathcal{E}_{K'}(t)\rangle \right), \quad (\text{D1})$$

where $|\mathcal{E}_K(t)\rangle$ and $|\mathcal{E}_{K'}(t)\rangle$ are the environmental states conditioned on the exciton valley, evolving from a common initial state $|\mathcal{E}_0\rangle$ at the moment of absorption ($t = 0$). The coherence between the two branches is encoded in the off-diagonal element of the density matrix obtained by tracing over the environment:

$$\rho_{K,K'}^{(\text{mat+ph})}(t) = \frac{e^{-i\varphi}}{2} \langle \mathcal{E}_K(t) | \mathcal{E}_{K'}(t) \rangle, \quad (\text{D2})$$

where $\rho_{K,K'}^{(\text{mat+ph})}$ denotes the coherence in the joint (material + idler photon) reduced state. The factor $\langle \mathcal{E}_K(t) | \mathcal{E}_{K'}(t) \rangle$ quantifies how distinguishable the environment has become at time t .

To model the time dependence of $\langle \mathcal{E}_K(t) | \mathcal{E}_{K'}(t) \rangle$, we treat the environment as a bath that acquires which-valley information through acoustic-phonon-mediated dephasing. In the Markovian limit, the environmental overlap satisfies

$$\frac{d}{dt} \langle \mathcal{E}_K(t) | \mathcal{E}_{K'}(t) \rangle = -\gamma_{\text{valley}} \langle \mathcal{E}_K(t) | \mathcal{E}_{K'}(t) \rangle, \quad (\text{D3})$$

where γ_{valley} is the rate at which the environment distinguishes the two valleys. This Markovian description is valid when the bath correlation time τ_c is much shorter than the timescale $1/\gamma_{\text{valley}}$ over which the decoherence factor evolves appreciably. For acoustic phonons in monolayer WSe₂, τ_c is set by the inverse of the acoustic-phonon bandwidth that couples to the exciton and is on the order of tens of femtoseconds, while $1/\gamma_{\text{valley}} \sim 1\text{--}3$ ps; the condition $\tau_c \ll 1/\gamma_{\text{valley}}$ is thus well satisfied. Integrating Eq. (D3) gives

$$\langle \mathcal{E}_K(t) | \mathcal{E}_{K'}(t) \rangle = e^{-\gamma_{\text{valley}} t}, \quad (\text{D4})$$

the standard decoherence-function decay from open quantum systems.

Before averaging the decoherence factor over dwell times, we need the probability distribution of those dwell times. Two ingredients from Appendix B set this distribution. First, the Fourier representation of the sinc joint spectral amplitude restricts τ to the causal interval $[0, T_e/2]$. Second, the retarded propagator for the intermediate exciton [Eq. (A2)] supplies an exponential weight $e^{-\gamma_X \tau}$.

The biexciton generation rate from intermediate dwell time τ is therefore proportional to $e^{-\gamma_X \tau}$ on $[0, T_e/2]$ and vanishes outside. Normalizing this weight to unit total probability defines the truncated-exponential dwell-time

distribution

$$p(\tau) = \frac{e^{-\gamma_X \tau}}{\int_0^{T_e/2} e^{-\gamma_X \tau'} d\tau'} = \frac{\gamma_X e^{-\gamma_X \tau}}{1 - e^{-\gamma_X T_e/2}}, \quad \tau \in [0, T_e/2]. \quad (\text{D5})$$

Equipped with $p(\tau)$, we now average the decoherence factor. Within the valley-symmetric limit $\mathcal{M}_2 = \mathcal{M}_3 \equiv \mathcal{T}$ [Eq. (26)], the joint biexciton + environment state at intermediate-exciton dwell time τ is

$$|\Psi(\tau)\rangle = \frac{\mathcal{T}}{\sqrt{2}} (|B_2\rangle |\mathcal{E}_K(\tau)\rangle + e^{i\varphi} |B_3\rangle |\mathcal{E}_{K'}(\tau)\rangle), \quad (\text{D6})$$

where $|\mathcal{E}_K(\tau)\rangle$ and $|\mathcal{E}_{K'}(\tau)\rangle$ are the environment states after dwelling for time τ with the intermediate exciton in valley K and K' , respectively. The biexciton reduced density matrix $\rho_{\text{BX}}(\tau) \equiv \text{Tr}_{\mathcal{E}}[|\Psi(\tau)\rangle \langle \Psi(\tau)|]$ then has configuration-basis matrix elements

$$\langle B_2 | \rho_{\text{BX}}(\tau) | B_2 \rangle = \langle B_3 | \rho_{\text{BX}}(\tau) | B_3 \rangle = \frac{|\mathcal{T}|^2}{2}, \quad (\text{D7a})$$

$$\begin{aligned} \langle B_2 | \rho_{\text{BX}}(\tau) | B_3 \rangle &= \frac{|\mathcal{T}|^2}{2} e^{-i\varphi} \langle \mathcal{E}_K(\tau) | \mathcal{E}_{K'}(\tau) \rangle \\ &= \frac{|\mathcal{T}|^2}{2} e^{-i\varphi} e^{-\gamma_{\text{valley}} \tau}, \end{aligned} \quad (\text{D7b})$$

where the final equality uses Eq. (D4). The diagonal populations are τ -independent, while the off-diagonal coherence carries the decoherence factor $e^{-\gamma_{\text{valley}} \tau}$ acquired during the dwell.

Averaging over τ with $p(\tau)$ [Eq. (D5)] therefore leaves the populations unchanged and suppresses the off-diagonal coherence by the dwell-time-averaged overlap, which we denote with an overline:

$$\begin{aligned} \overline{\langle \mathcal{E}_K | \mathcal{E}_{K'} \rangle} &\equiv \int_0^{T_e/2} p(\tau) \langle \mathcal{E}_K(\tau) | \mathcal{E}_{K'}(\tau) \rangle d\tau \\ &= \int_0^{T_e/2} p(\tau) e^{-\gamma_{\text{valley}} \tau} d\tau \\ &= \frac{\gamma_X}{\gamma_X + \gamma_{\text{valley}}} \frac{1 - e^{-(\gamma_X + \gamma_{\text{valley}}) T_e/2}}{1 - e^{-\gamma_X T_e/2}}, \end{aligned} \quad (\text{D8})$$

Because the diagonal populations carry no φ dependence while the off-diagonal coherence carries $e^{-i\varphi} \overline{\langle \mathcal{E}_K | \mathcal{E}_{K'} \rangle}$, feeding $\overline{\rho_{\text{BX}}}$ into any bright-eigenstate pumping rate produces a φ -scan intensity $I(\varphi) \propto 1 + \overline{\langle \mathcal{E}_K | \mathcal{E}_{K'} \rangle} \cos \varphi$. The dwell-time-averaged overlap is therefore the fringe visibility, $V_{\text{deph}} = \overline{\langle \mathcal{E}_K | \mathcal{E}_{K'} \rangle}$, giving Eq. (47).

The same distribution gives the mean intermediate-state dwell time:

$$\begin{aligned} \langle \tau \rangle &\equiv \int_0^{T_e/2} \tau p(\tau) d\tau = \frac{\int_0^{T_e/2} \tau e^{-\gamma_X \tau} d\tau}{\int_0^{T_e/2} e^{-\gamma_X \tau} d\tau} \\ &= \frac{1}{\gamma_X} - \frac{T_e/2}{e^{\gamma_X T_e/2} - 1}, \end{aligned} \quad (\text{D9})$$

which is Eq. (46). In the narrow-bandwidth limit ($\gamma_X T_e \gg 1$), the second term vanishes and $\langle \tau \rangle \rightarrow 1/\gamma_X$. In the broad-bandwidth limit ($\gamma_X T_e \ll 1$), expanding the exponential to second order gives $\langle \tau \rangle \rightarrow T_e/4$. This mean dwell time sets the exposure time of the intermediate exciton to the environment and enters the Sec. III B visibility through the averaged overlap V_{deph} [Eqs. (47) and (D8)].

2. Combined visibility from intervalley scattering

Unlike valley dephasing, which degrades the off-diagonal coherence without changing the pathway population, intervalley scattering physically transfers the intermediate exciton from one valley to the other at rate γ_{iv} . An exciton created in valley K at dwell time $\tau = 0$ survives in K with probability $e^{-\gamma_{\text{iv}}\tau}$. Averaging this survival probability over the same dwell-time distribution $p(\tau)$ [Eq. (D5)] used for the valley-dephasing calculation gives

$$\begin{aligned} \overline{e^{-\gamma_{\text{iv}}\tau}} &= \int_0^{T_e/2} p(\tau) e^{-\gamma_{\text{iv}}\tau} d\tau \\ &= \frac{\gamma_X}{\gamma_X + \gamma_{\text{iv}}} \frac{1 - e^{-(\gamma_X + \gamma_{\text{iv}})T_e/2}}{1 - e^{-\gamma_X T_e/2}}, \end{aligned} \quad (\text{D10})$$

which is identical in form to Eq. (D8) with the replacement $\gamma_{\text{valley}} \rightarrow \gamma_{\text{iv}}$. The fraction of each pathway that is scattered into the wrong valley is therefore

$$P_{\text{iv}} = 1 - \overline{e^{-\gamma_{\text{iv}}\tau}} = 1 - \frac{\gamma_X}{\gamma_X + \gamma_{\text{iv}}} \frac{1 - e^{-(\gamma_X + \gamma_{\text{iv}})T_e/2}}{1 - e^{-\gamma_X T_e/2}}, \quad (\text{D11})$$

which is Eq. (48).

The scattered fraction converts each pathway into the wrong valley:

$$|X_K\rangle \rightarrow \sqrt{1 - P_{\text{iv}}} |X_K\rangle + \sqrt{P_{\text{iv}}} e^{i\chi_K} |X_{K'}\rangle, \quad (\text{D12a})$$

$$|X_{K'}\rangle \rightarrow \sqrt{1 - P_{\text{iv}}} |X_{K'}\rangle + \sqrt{P_{\text{iv}}} e^{i\chi_{K'}} |X_K\rangle, \quad (\text{D12b})$$

where $\chi_K, \chi_{K'}$ are random phases associated with the scattering event. After the second photon is absorbed, the (σ^+, σ^-) pathway produces $|B_2\rangle$ with probability $1 - P_{\text{iv}}$ and the corrupted co-circular configuration $|B_6\rangle$ with probability P_{iv} ; similarly the (σ^-, σ^+) pathway produces $|B_3\rangle$ or the corrupted configuration $|B_1\rangle$. The corrupted configurations do not interfere with the cross-circular sector $\{|B_2\rangle, |B_3\rangle\}$, so they add a phase-independent background without affecting the cross-circular coherence.

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